

Captured by the Words: Literal Attraction in Typed Decision Models Tracks the Test Language Rather Than Capability on Bangla Idioms

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Abstract

Decision models, non-generative scorers that report a probability over caller-supplied options and emit no text, are increasingly deployed as routers, classifiers and guardrails. Because they generate nothing, rationales and chain-of-thought do not apply, and idioms make token attribution hard to interpret: for an expression whose figurative reading is not recoverable from its parts, it is unclear what a per-word contribution would mean. We therefore audit three such models, an encoder scorer, a decoder scorer and a closed commercial API, entirely through interventions on the option set, using expert-annotated Bangla idioms that ship with both a gold figurative gloss and a gold literal gloss. Dropping an idiom’s own literal gloss into the option list captures all three models. Running the full model $\times$ language matrix over one shared item set shows the failure tracks the test language rather than capability: every arm is several times more captured in Bangla than in English on the same paired entries, and the commercial model’s apparent resistance turns out to be a property of the language it was tested in. Confidence inverts under the same shift, so gating on it does worse than gating at random, and on a semantics-free option list, where no option carries information and expected accuracy is 0.25, the best-calibrated arm still reports 0.82 confidence. Option-order sensitivity, by contrast, separates the two scoring architectures rather than the three models.

1 Introduction

A *decision model* is given a question and a list of candidate options and returns a probability distribution over exactly those options. It is not autoregressive, it does not sample, and it produces no text. This class has grown quickly as infrastructure for routing, intent classification, content gating and tool selection, precisely because it is cheap,

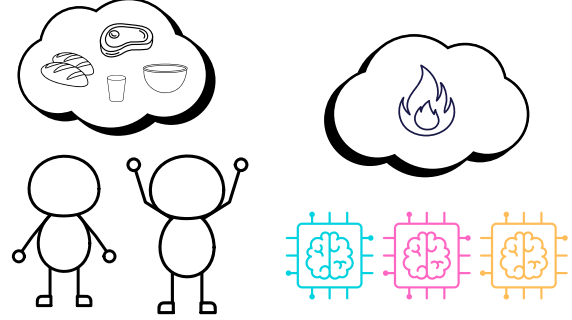


Figure 1: **The model takes the words.** *Laya-ml*, *Lod-lille*, and *Jev-1.13* are shown in cyan, magenta, and orange, respectively, throughout the paper. The Bangla idiom *অগ্নিবল* (Agnibal) is composed of *অগ্নি* (*fire*) and *বল* (*strength*), but means intense hunger; English expresses the same concept as *fire in the belly*. When both readings are placed in the option list, all three models choose the literal gloss in Bangla, at $p = 1.00, 0.97$, and 0.36 , respectively; with the same four options in English, two of the three recover. Across 3,265 paired items, literal attraction falls from 0.731, 0.918, and 0.304 in Bangla to 0.379, 0.241, and 0.047 in English (§5.1).

bounded and returns a number a caller can threshold on. An ecosystem analysis of 2,170 public projects finds it already embedded as a reusable decision component across domains (Ling et al., 2026), and proposals now place these models at the adjudication points of autonomous security harnesses (dos Santos Barbosa, 2026). The threshold is load-bearing.

That bounded interface is also an explainability problem. There is no rationale to inspect, no chain of thought to measure for faithfulness (Jacovi and Goldberg, 2020; Turpin et al., 2023), and in the commercial case no weights at all. What is left is the behavioral surface: the caller controls the question and the option list, and sees a probability vector. Any audit of this class has to be built out of that.

The obvious fallback, attributing the decision to input tokens (Ribeiro et al., 2016; Lundberg and Lee, 2017), is both unreliable here (Jain and Wal-

lace, 2019; Adebayo et al., 2018; Hooker et al., 2019) and hard to read. How far an idiom’s parts map onto components of its figurative meaning varies by idiom and is not annotated in the corpus we use (Nunberg et al., 1994; Gibbs et al., 1989); for an expression at the non-decomposable end, a Shapley value over its words is computable but has no evident linguistic referent. We would rather intervene on the expression as a unit than report shares we cannot interpret (Appendix D).

We take the option set as the intervention surface instead. Bangla *bāgdhārā* (idioms) are well suited to this because the expert resource we use ships, for each idiom, both a gold figurative meaning and a gold *literal* gloss: a ready-made, human-certified counterfactual distractor that no annotation effort of ours produced. Dropping that gloss into the option list turns a benchmark question into a targeted probe: does the model read the expression as an idiom, or as its words? Figure 1 shows one item answered both ways.

We contribute (i) a meaning-choice instrument for decision models in which the *distractor family* is the independent variable and everything else is held fixed, requiring no new annotation (§4); (ii) the Literal Attraction Rate, measured as a full model $\times$ language matrix, which shows capture tracking the test language rather than capability (§5.1); (iii) evidence that confidence is anti-diagnostic under this shift across three different calibration philosophies, with confidence-gated selective prediction worse than random gating (§5.2); (iv) a large dissociation in option-order dependence between the two scoring designs, with a methodological consequence for anyone evaluating this class (§5.4); and (v) a semantics-free control, the one condition that runs identically on all three arms, bounding what each model reports when the options carry no information at all (§5.3).

2 Related Work

The decision-model class. The evaluation literature is still forming, and reports competitive quality at far lower cost with no independent accuracy advantage from the typed readout itself (Tang and Zheng, 2026; Rao and Callison-Burch, 2026; Ibrahim and Zaki, 2026; Guo et al., 2026; Yu and Yao, 2026). Two strands bear on ours. Ibrahim and Zaki (2026) report decision-model *calibration exceeding most LLM baselines*, and Li et al. (2026) build the confidence-gated cascade that follows

from it; both depend on the returned confidence being diagnostic, and §5.2–§5.3 give conditions under which it is not. Concurrently Sun et al. (2026) find the decision follows an option’s *name* rather than its bound definition, and Xu (2026) flip 61.4% of correct decisions with short, fluent context additions chosen to redirect them. Appendix B sets out how each relates to our design.

Idioms in NLP. Idiom resources and tasks are well established (Haagsma et al., 2020; Tayyar Madabushi et al., 2022; Chakrabarty et al., 2022), including for Bangla (Sakhawat et al., 2026), and the difficulty of figurative language for distributional models is long-noted (Shwartz and Dagan, 2019). Psycholinguistics frames the phenomenon as competition between a literal and a figurative route whose resolution depends on decomposability and context (Cacciari and Tabossi, 1988; Gibbs et al., 1989; Titone and Connine, 1999), and Oh et al. (2026) give mechanistic evidence that the competition is real inside a transformer: causal tracing recovers two pathways carrying the figurative and literal readings simultaneously, with early layers promoting the figurative one and a direct route keeping the literal one available.

Our D-LIT intervention is the behavioral counterpart of that picture rather than a replication of it: they can ask *how* the literal route stays available inside a model they can open, we can only ask *how often* it wins in one that emits no text and, in one arm, exposes no weights (Appendix B).

Evaluating option scorers. Option-order sensitivity and selection bias are documented for prompted generative LMs (Pezeshkpour and Hruschka, 2024; Wei et al., 2024; Zheng et al., 2024; Zhao et al., 2021). We are not aware of a comparison of option-marker against independent-option scoring in small non-generative scorers over the full permutation set, which is what §5.4 provides.

Calibration and selective prediction. Proper scoring and calibration measurement are standard (Brier, 1950; Gneiting and Raftery, 2007; Guo et al., 2017; Nixon et al., 2019; Kumar et al., 2019), as is the observation that calibration degrades under distribution shift (Ovadia et al., 2019; Jiang et al., 2021). Selective prediction gives the operational test (Geifman and El-Yaniv, 2017, 2019; Traub et al., 2024). We report the combination that matters for deployment: accuracy *decreasing* with confidence, and a confidence gate underperform-

ing a random one.

Behavioral testing. Our design follows behavioral checklists, contrast sets and counterfactual minimal pairs (Ribeiro et al., 2020; Gardner et al., 2020; Kaushik et al., 2020; Naik et al., 2018; Jia and Liang, 2017); the contribution is carrying that tradition into a model class with no text output, where it is one of the few instruments left.

Low-resource evaluation. Bangla has strong encoders and corpora (Bhattacharjee et al., 2022; Kakwani et al., 2020), and cross-lingual benchmarks establish the transfer gap (Hu et al., 2020; Ruder et al., 2021; Adelani et al., 2021); §5.7 localizes our failure to the Bangla side rather than to figurativeness.

3 Dataset

Source corpus. We use *Bangla Bagdhara* (Sakhawat et al., 2026), a corpus of 10,361 Bangla (Bengali) idioms annotated under a 19-field schema by expert consensus.¹ Each row carries the idiom, a gold figurative meaning in Bangla, an expert literal gloss, semantic tags, alternative surface forms certified as meaning-preserving, and a gold carrier sentence. We add no annotation; every intervention below is assembled from fields that already exist.

Its own benchmark on the same underlying task reports no model among 30 multilingual and instruction-tuned LLMs above 50% accuracy, against 83.4% for human annotators. Option construction differs, so those figures are not directly comparable to ours, but they locate the difficulty in the material rather than in our instrument.

4 Methodology

Figure 2 sets out the instrument end to end: what the corpus supplies, what each condition changes, and what comes back.

4.1 The meaning-choice task

Given an idiom, choose its gold figurative meaning from $k = 4$ candidates. Chance is 0.250 throughout. The **distractor family is the independent variable**; the item, the instruction, k , and per-item option-order randomization (seeded by item id) are held fixed across families, so a paired test across families is a test on byte-identical questions.

¹Two rows are lost to parsing, leaving 10,359; of these 8,876 carry every field the instrument requires.

4.2 Distractor families and measures

The five families are D-RAND (other idioms’ glosses at random), D-SURF (glosses sharing surface tokens with the gold), D-SEM (glosses from idioms sharing semantic tags), D-LIT (one distractor replaced by the idiom’s *own* expert literal gloss) and D-HARD (all three pressures at once). Appendix C gives each sampling rule, and Table 6 one complete item in the original script.

For the families containing a literal gloss, with ℓ_i the literal option and g_i the gold, the Literal Attraction Rate is

$$\text{LAR} = \frac{1}{N} \sum_{i=1}^N \mathbf{1} \left[\arg \max_o p_i(o) = \ell_i \right].$$

LAR is not $1 - \text{accuracy}$: it isolates the errors that are specifically a literal reading, and has no floor, since a model that never reads literally scores 0 while still being wrong.

4.3 Interventions and statistics

Seven interventions vary one thing each: the distractor family (E1), the idiom’s surface form (E2), the idiom as a unit (E3), the task language (E4), nothing (E5, derived from E1), the option text (E6) and the option order (E7). Appendices C and A give what each holds fixed and the typed question they share.

All intervals are 95% percentile bootstrap over items (2,000 resamples); paired comparisons use McNemar’s exact test; ECE uses 15 equal-width bins and AUC is reported against a random-gate baseline. The two open arms were built independently from a written contract, and on all 4,440 overlapping rows the gold key, gold position, idiom string and literal-distractor key agree at 1.000, so every paired test is on identical questions (Appendix R).

4.4 Models

Three models span the design space (Table 1); none has an accompanying paper, so we identify them by their released artifacts.² **Laya-ml** is a bidirectional encoder scoring all options together from marker positions. **Lod-lille** is a decoder with

²**Laya-ml:** convaiinnovations/laya, subfolder multilingual, on the Hugging Face Hub; the same repository ships the English-only checkpoint used as the control in §5.7. **Lod-lille:** mrn-dk/lod-lille-0.6B, same hub. **Jev-1.13:** typesafe/jev-1.13, served over OpenRouter’s /api/alpha/decisions endpoint, with no weights, training data or architecture published.

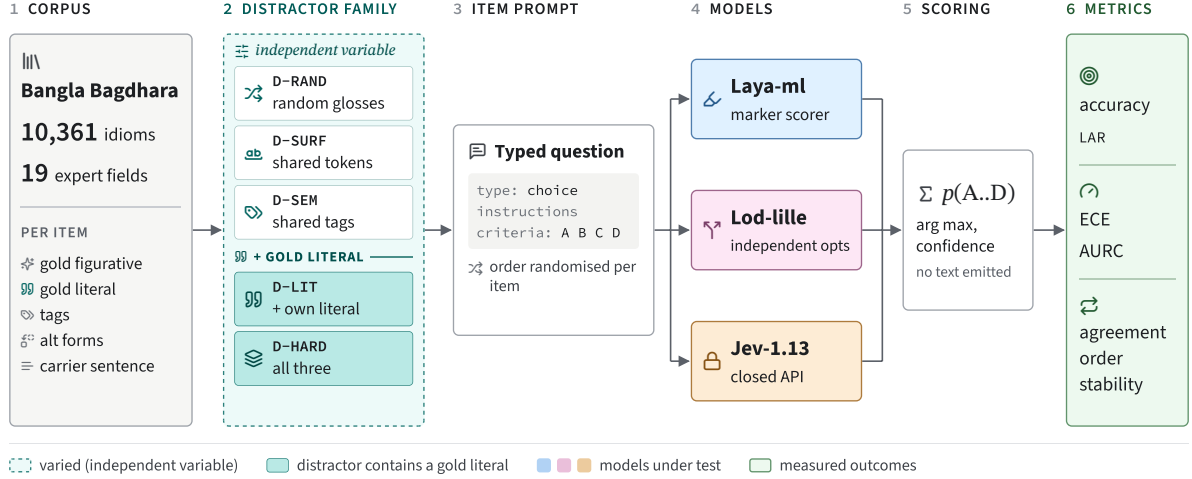


Figure 2: **The instrument in one view.** The audit proceeds from the **Bangla Bagdhara** corpus through six stages: (1) **corpus**, which supplies 10,361 expert-annotated idioms with gold figurative and literal glosses; (2) **distractor family**, the sole manipulated variable, with D-LIT and D-HARD injecting the expert literal gloss; (3) **item prompt**, which fixes the typed question, $k = 4$ options, instructions, and per-item order seed; (4) **models**, comprising Laya-ml, Lod-lille, and Jev-1.13; (5) **scoring**, which records the four-option probability vector, arg max, and confidence without requiring generated text; and (6) **metrics**, which evaluate accuracy, LAR, ECE, AURC, agreement, and order stability. The figure’s visual encoding distinguishes the varied distractor family, literal-bearing conditions, models under test, and measured outcomes.

	Laya-ml	Lod-lille	Jev-1.13
backbone	mmBERT	Qwen3-0.6B	closed
scoring	marker	independent	unknown
parameters	322M	0.6B	n/a
items	8,876	888	8,761
latency	27.2 ms	86.1 ms	API
weights	open	open	closed

Table 1: **The three model arms.** Laya-ml is a 322M-parameter mmBERT marker scorer, Lod-lille is a 0.6B-parameter Qwen3-based independent-option scorer, and Jev-1.13 is a closed API with an unknown scoring architecture. The item counts and latencies are those used in the reported experiments. Lod-lille was run on a fixed 10% random subset; Appendix O bounds the resulting drift at 0.016.

Distractors	Laya	Lod	Δ
random	0.367	0.484	+0.133***
surface	0.338	0.412	+0.073***
tag	0.333	0.446	+0.120***
+ literal	0.151	0.066	-0.070***
hard	0.144	0.063	-0.086***
LAR (lit.)	0.644	0.885	
LAR (hard)	0.630	0.855	
mean p (lit.)	0.464	0.726	
mean p (gold)	0.195	0.128	

Table 2: **The distractor ladder and literal attraction.** McNemar tests use the 888 shared items (***) $p < 0.001$. Cyan marks Laya-ml, while magenta marks Lod-lille; darker cyan and magenta fills high-light the higher value within a row. The sign of Δ reverses at the literal boundary: the stronger arm leads on the first three families and trails on the two that contain a literal gloss.

independent-option scoring, each option attending only to the shared state, the question and itself from the same start position, which its model card reports as order-independent by construction in fp32; §5.4 finds that it is not. **Jev-1.13** is a closed commercial model behind a decisions API whose documentation names English as its primary and best-supported language; we know its interface and nothing else. Appendix A gives the call each one receives.

5 Analysis

5.1 Literal capture, and what it tracks

Table 2 and Figure 3 give the ladder for the two open arms, each on its own item set, with the Δ

column paired on the 888 items they share; Appendices E and G give all three arms and the stratifications. Lod-lille is the stronger model on every non-literal condition, by up to +0.133. Insert the idiom’s own literal gloss and the ordering *reverses*: Lod-lille falls to 0.066, less than half of Laya-ml’s already-below-chance 0.151, and the same holds for D-HARD.

Within this pair, then, LAR measures something accuracy does not: a model can be selected as “better” on a conventional benchmark while being worse at what that benchmark proxies for. Laya-ml

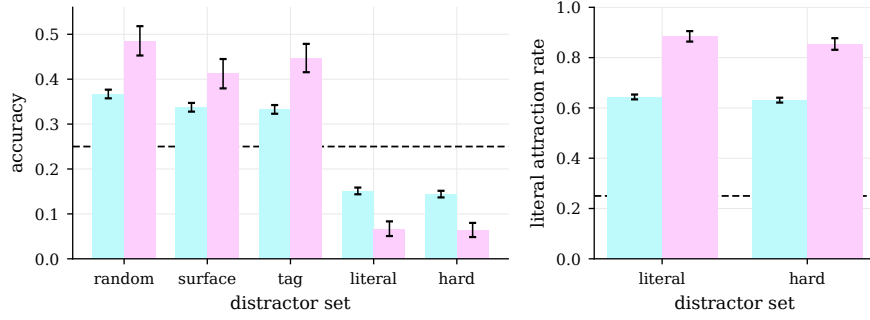


Figure 3: The distractor ladder (left) and literal attraction rate (right) for the two open arms. ■ Laya-ml ($n = 8,876$), ■ Lod-lille ($n = 888$); the black dashed line marks the four-option chance level (0.250) in both panels; error bars are 95% bootstrap CIs. Surface- and tag-matched distractors cost a few points; the idiom’s own literal gloss (D-LIT) collapses both models far below chance, and the *stronger* model harder.

and Lod-lille answer byte-identical questions in the same language, so the crossover between them is a paired result. The error is not diffuse: the decomposition under D-LIT is gold / literal / other = 0.151 / 0.644 / 0.207 for Laya-ml and 0.066 / 0.885 / 0.050 for Lod-lille, and on the clearest items Laya-ml places $p = 1.000$ on the literal gloss.

Capture tracks the language, not capability.

The arms above ran the literal condition in Bangla, while the commercial arm’s headline $LAR = 0.047$ is *English*, so the two are not comparable as they stand. We therefore ran the full model $\times$ language matrix on the 3,265-item shared pool, every cell built by one function from one seed (Table 15, Figure 4); Jev-1.13’s English cell reproduces its published 0.0472 [0.0401, 0.0545], which checks that the two constructions measure the same thing.

Two things follow. First, **every arm is far more captured in Bangla than in English on the same paired entries**: $0.731 \rightarrow 0.379$, $0.918 \rightarrow 0.241$ and $0.304 \rightarrow 0.047$, all paired McNemar $p < 10^{-100}$, so the commercial model’s apparent resistance was a property of its test language. Second, **capture is not monotone in capability**: Lod-lille is more accurate than Laya-ml and more captured, Jev-1.13 more accurate than both and less captured. The open arms fall below chance here; Jev-1.13 does not, but loses 0.390 of accuracy against its own English run. The pool is not neutral and its Bangla literal option carries a cue the English one lacks; on the 5,485 corpus items with no such cue, Bangla capture is still 0.581 and 0.871 on the open arms (Appendices I, J).

The capability ordering. One condition is identical across all three arms, Bangla idiom with random distractors, and on it the commercial model is far ahead, 0.736 against 0.367 and 0.484 (Ap-

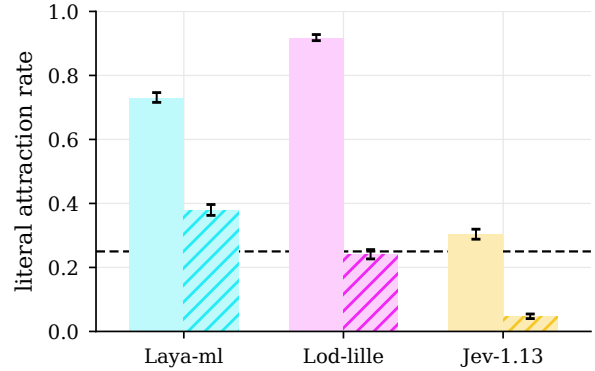


Figure 4: Literal attraction by model and task language, all six cells on the same 3,265 items (error bars 95% bootstrap CIs; dashed line the four-option chance level). ■ Laya-ml, ■ Lod-lille, ■ Jev-1.13; solid bars Bangla, hatched bars English. The Bangla ordering does not follow capability, and Jev-1.13’s English value reproduces the 0.047 reported from its original run. Cells with intervals in Table 15.

pendix L). That is the ordering the calibration results below should be read against, and the one capture does not follow.

5.2 Confidence is anti-diagnostic

Table 3 gives the mean probability on the chosen option, split by outcome, on the literal-bearing families. Both open models are *more confident when captured than when correct*, Lod-lille extremely so (0.787 against 0.453), and their reliability curves slope *downward* under D-LIT (Figure 5).

ECE depends on the binning (Nixon et al., 2019), so we also report the multiclass Brier score, which needs none: it is 1.030 and 1.391 for the two open arms under D-LIT against 0.750 for a uniform four-way distribution, so their reported probabilities are worse than declining to choose. Equal-mass bins move every ECE by less than 0.011 (Appendix F).

Selective prediction therefore inverts: restricting Laya-ml’s D-LIT to its most confident 5% gives

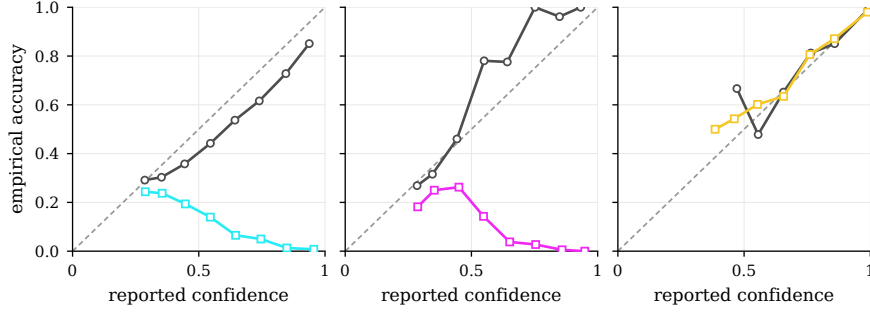


Figure 5: Reliability, 15 equal-width bins. Panels left to right: Laya-ml, Lod-lille, Jev-1.13. ■ circles are the random-distractor condition, squares in each panel’s own color the +literal one, and the gray dashed diagonal perfect calibration. Under D-LIT the two open models’ curves bend *downward*: the model is most wrong where it reports being most sure. Jev-1.13, in English, stays on the diagonal.

Outcome	Laya	Lod	Jev
correct	0.432	0.453	0.935
captured	0.596	0.787	0.737
other error	0.421	0.399	0.746
ECE (D-RAND)	0.071	0.073	0.005
ECE (D-LIT)	0.385	0.679	0.014
AURC (conf.)	0.924	0.989	0.013
AURC (random)	0.849	0.934	0.069

Table 3: Confidence by outcome on the literal-bearing families, and calibration. Bold marks, per model, whether it is more confident when correct or when captured. Both open arms are more confident when captured, and their confidence-gated AURC is *worse* than the random-gate baseline in the gray row, which inverts selective prediction. Jev-1.13’s English condition does not show the inversion. Appendix F adds Brier and equal-mass ACE for the same cells.

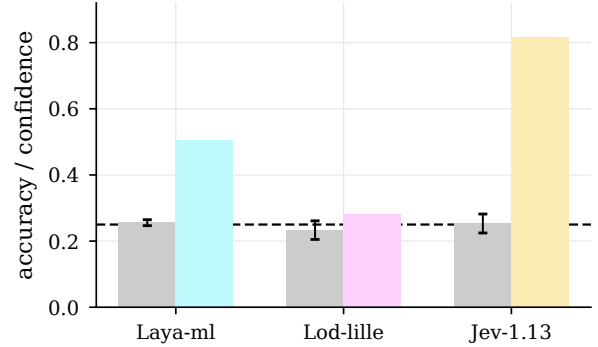


Figure 6: The semantics-free control. Per model, the ■ bar is accuracy (95% bootstrap CI) and the colored bar is mean reported confidence (Laya-ml, Lod-lille, Jev-1.13); the dashed line is the four-option chance level. No option carries information, so the gap between the two bars is overconfidence: 0.250, 0.049 and 0.561. Jev-1.13’s condition is English.

0.007 accuracy against 0.151 at full coverage, and Lod-lille answers every item in its most confident 35% incorrectly (Appendix F). This is not an artifact of one calibration philosophy: Laya-ml ships fitted per-bucket temperatures, Lod-lille ships none by design, and Lod-lille’s *separate confidence head*, meant to estimate whether the answer is in the option list at all, shows the same inversion (0.631 on captured items against 0.295 on correct ones).

Jev-1.13 is the exception, and only in one direction. Its English reliability is close to ideal (ECE 0.005 and 0.014, rising to 0.047 on D-HARD) and its confidence-gated AURC beats random by an order of magnitude, which makes §5.3 harder to dismiss rather than easier. Our claim is the narrow one, and it does not contradict Ibrahim and Zaki (2026): in-distribution calibration need not survive a targeted distractor shift, and it fails in the costly direction, since a confidence-gated cascade (Li et al., 2026) accepts without review exactly the tail where the open arms are *least* accurate (Appendix B).

5.3 A semantics-free control, on all three arms

The o4 ablation replaces every option’s text with a bare label. With the gold position randomized and no option carrying information, expected accuracy is 0.25 and whatever confidence survives is overconfidence, with no modeling assumption needed to read it. It is also the only condition that runs identically on all three arms, needing no gloss, no literal annotation and no shared language, so the language confound scoping §5.1–§5.2 to the open arms does not arise (Figure 6).

The ordering reverses against Table 3. The best-calibrated model under real options is the worst when the options carry nothing: Jev-1.13 reports 0.815 mean confidence at 0.254 accuracy, an overconfidence of 0.561, which we give as a gap rather than a binned ECE because the gap is a lower bound on ECE and needs no binning choice to defend. Lod-lille, the weakest of the three on accuracy, is the only one whose confidence collapses appropriately (0.282). We read this as the most

	Laya	Lod	Jev
answer changes	0.837	0.107	0.133
correctness varies	0.686	0.070	0.120
distinct glosses	2.83	1.14	1.15
perfectly stable	0.163	0.893	0.867
$p(\text{gold})$ range	0.346	0.029	0.128
selection spread	0.139	0.030	0.011

Table 4: **Option-order sensitivity.** Each item is evaluated under all 24 orderings of the same four options ($n = 992/299/300$). Laya, Lod, Jev; bold marks the largest value in each row. The marker-based scorer changes its chosen gloss on 83.7% of items against 10.7% for the independent-option scorer. Lod is documented as order-independent by construction; the audit in Appendix Q shows it is not, which is why its entries here are small rather than zero.

portable finding here, because it needs no annotation and no matched language to reproduce: a calibration number measured on in-distribution options says little about what a model reports when the option list is degenerate, and a degenerate option list is an ordinary consequence of an upstream bug or an empty retrieval rather than an adversarial construction (Appendix B). McNemar between the two open arms under o4 is not significant ($p = 0.76$), so the instrument’s null behaves alike in both.

5.4 Option order is architecture-specific

We run every item under all $4! = 24$ orderings of a fixed option set and record the identity of the chosen gloss, recovered as $\text{perm}[\text{slot}(\hat{y})]$; Appendix Q explains why the slot letter is the wrong unit here.

Table 4 gives a large dissociation: a $7.8\times$ difference in answer stability and a $12\times$ difference in probability stability. Each scoring design is represented by exactly one model here, so the design cannot be separated from the backbone or the capability of the model carrying it; we read the pattern as consistent with the scoring architecture rather than as demonstrated to follow from it.

The invariance claim does not hold. Lod-lille is documented as order-independent by construction, each option being scored in its own forward pass. Mapping every slot’s probability back to the option it displayed, a single option’s own score travels 4.0×10^{-2} at the median item and never stays below 10^{-4} , orders of magnitude above fp32 noise (Appendix Q). The design attenuates order dependence by roughly an order of magnitude without removing it, so the guarantee is not one an evaluation could rest on.

Positional bias follows. We measure it on the full ladder run rather than on this sweep, since there the gold position is randomized per item across all 8,876: Laya-ml takes slot B 37.2% of the time against 14.6% for slot A (Appendix Q).

Any evaluation of this class that fixes option order is therefore partly measuring position. All results here randomize order per item, seeded by item id, and we would treat that as a default for this class rather than a robustness check.

5.5 Options are read as a bag of words

The o-series ablates the option text while holding the idiom fixed (Figure 10, Appendix P; Table 11 shows the forms applied to real Bangla). Truncating each option to six words and *scrambling the words inside it* cost almost nothing. Writing *signal* for accuracy above chance as a fraction of the full-gloss condition’s, $(a - 0.25)/(a_{o0} - 0.25)$, Laya-ml retains 0.969 and 0.928 of its signal, Lod-lille 0.902 and 0.944, and Jev-1.13 is flat to within noise. Behaviorally, what all three do on the option side is close to order-free lexical matching, which converges with Sun et al. (2026) from the other direction (Appendix B). The drop to the gray semantics-free bar is the only large one, which certifies the instrument: the task measures option semantics, as a bag.

5.6 Neither open model treats the idiom as a unit

E3 perturbs the idiom while holding the option list fixed (Figure 9, Appendix M). It stands in for token attribution: rather than asking what each word contributes, which presumes a decomposition the expression may not support, we ask whether the model behaves as though it stores the expression as a unit, since a model that did would lose the figurative reading when the unit is broken.

Neither does. Word-shuffling leaves Laya-ml’s accuracy statistically unchanged ($0.157 \rightarrow 0.168$) and reversal helps slightly (0.178); Lod-lille moves within noise. Agreement with the intact answer under shuffling is 0.742 and 0.917, so Lod-lille is more *stable* but no more *compositional* (Table 11).

The one manipulation that clearly helps both is embedding the idiom in its gold carrier sentence, $+0.080$ and $+0.117$: context, not the idiom string, carries the usable signal. That cuts both ways, since Xu (2026) flip 61.4% of initially correct decisions with short, fluent additions chosen to redirect them, and nothing in the interface dis-

tinguishes a helpful addition from a harmful one (Appendices M, B).

5.7 The failure localizes to Bangla, in every arm

Holding the model fixed and swapping the task language, now on random distractors rather than the literal family (Figure 7, Appendix K), moves every arm by a large margin: Laya-ml +0.246, Lod-lille +0.380, Jev-1.13 +0.230, each paired on that arm’s own shared items (all McNemar $p < 0.001$). Lod-lille reaches 0.864 in English against 0.484 in Bangla, so the figurative concepts appear largely available to it and the Bangla route is where the loss occurs, which is consistent with its documented English-only training evaluation. Laya-ml’s control supports the reading: its English-only checkpoint on Bengali script sits at 0.283, barely above chance, making it unlikely the Bangla result is an artifact of the scorer rather than of the language. Jev-1.13 complicates it and then confirms it, outscoring both multilingual arms on the Bangla task while its own capture still rises from 0.047 to 0.304 there. Being better at a language is not the same as being robust in it. Here too the English condition presents the annotator’s nearest English idiom rather than a translation, so these are paired entries (Appendix J).

6 Discussion

The matrix in §5.1 is the result we would defend hardest, because it separates two explanations that are easy to conflate. Read only in English, the commercial arm looks immune to literal capture; read in Bangla it is captured at six times the rate. What a decision model does with a figurative expression is therefore not a fixed property of the model, and a benchmark run in one language will not tell you how it behaves in another.

The other results sit at different scopes. The semantics-free control runs identically on all three arms and needs no annotation or shared language, so it is the one we would most expect to transfer; the confidence inversion is a paired result on the two open arms. Option-order dependence separates the two scoring designs by roughly an order of magnitude without vindicating either, since the design documented as order-independent is not, by a margin far above numerical noise.

For practitioners we would draw three cautions rather than recommendations: a returned probabil-

ity is worth little without a measurement of what the model returns on a degenerate option list; a benchmark score is not by itself evidence against literal capture, since on our open arms the two move in opposite directions; and an evaluation that fixes option order is partly reporting position. None of this rebuts the cascade designs, whose cost and latency advantages are established (Tang and Zheng, 2026; Rao and Callison-Burch, 2026); it bounds them.

7 Conclusion

Decision models are being placed at decision points, and the interface that makes them attractive is also what makes them hard to audit: no rationale, no chain of thought, and in the commercial case no weights. We have argued that the option list is an adequate substitute: every result here comes from changing what is in that list and reading the probability vector back, which needs no new annotation and works unchanged on a closed, text-free model.

One substituted option turns out to be enough. Replacing a single random distractor with the idiom’s own expert literal gloss captures all three arms, at six to eleven times the cost of the hardest lexical or semantic distractor. The model $\times$ language matrix then separates two explanations that are easy to conflate: capture falls by 0.35 to 0.68 in every arm when the paired entries are asked in English, so the commercial model’s apparent resistance belongs to its test language rather than to the model. Capability does not order the failure either, the strongest arm being the least captured and the middle arm the most, so a benchmark ranking does not predict it.

The caller is not warned. Confidence inverts under the same shift, so a confidence gate performs worse than a random one, and on an option list stripped of semantics the arms still report between 0.28 and 0.82. Option order separates the two scoring designs by roughly an order of magnitude, though the design documented as order-independent turns out not to be. None of this is visible to a caller who checks only that the output validates. We bound the two construction cues our own design introduces and capture survives without them, but bounding is not removing, and one corpus with two small open models and a mutable endpoint is a narrow base. We release the instrument and the per-item outputs, and would most like to see this audit pointed elsewhere.

Limitations

All three models are evaluated zero-shot, and this is the limitation we would weigh most heavily after the one below. Decision models are trained on a particular set of decision tasks, and the open arms’ own documentation reports that base checkpoints are weak on tasks outside that set. Some part of what we measure may therefore be out-of-distribution task transfer rather than literal capture specifically. The o4 control bounds this from one side, since it shows the instrument’s null behaving identically in both open arms, and the D-RAND to D-LIT contrast is within-model and within-item, which removes any fixed task-transfer offset. Neither argument rules the confound out. A single lightly fine-tuned arm would settle it, and we have not run one.

Laya-ml’s run had its per-bucket temperatures unset, so its ECE figures are raw. Lod-lille ships no temperature by design, so both open arms are effectively untempered and comparable on that axis; Jev-1.13’s internal calibration is unknown to us.

Lod-lille ran a fixed 10% subset (888 items), so its intervals are roughly ± 3 points and its stratified cells are thin. Appendix O bounds the subset bias at 0.016 and shows it changes no sign or ordering. E3 on the Lod-lille arm is 180 items and several of its perturbation contrasts are not individually significant. E7 covers 992 / 299 / 300 items rather than the full sets.

The matrix in §5.1 runs on the 3,265 items that carry a usable English literal gloss, which is not a neutral sample: for Laya-ml, in-pool LAR is higher than out-of-pool while D-RAND accuracy is unchanged, so the absolute rates there are higher than they would be corpus wide. That difference is not generic pool selection. Pool membership requires an English parenthetical inside the Bangla literal gloss, and Appendix J shows the parenthetical is itself worth +0.165 of LAR on Laya-ml, which is most of the in-pool elevation. The language comparison is paired within item and within model, so the selection affects the levels rather than the direction. The matrix also covers the literal condition only; we did not run the other distractor families in both languages. The confidence results in §5.2 remain two-arm, since the commercial arm’s reliability was measured on its English run.

The two language cells are also not symmetric in construction, and this is the limitation we would flag hardest. They are paired entries rather than

identical items: the Bangla state is the idiom, while the English state is the annotator’s nearest English idiom rather than a translation of it. The two literal glosses are drawn from one annotated field in different ways, which leaves the Bangla literal option both lexically closer to its own state and, because it retains its English parenthetical, the only option in the list written partly in Latin script. Appendix J sets out all four asymmetries we could find and tests the two that recorded outputs can reach. Restricted to the 1,429 items whose Bangla literal gloss shares no content word with the idiom, the capture gap is +0.358, +0.610 and +0.310 against headline gaps of +0.352, +0.677 and +0.256. Restricted to the 5,485 corpus items whose literal gloss carries no English parenthetical at all, Bangla capture is still 0.581 and 0.871 on the two open arms, against a chance level of 0.250. Neither cue accounts for the result, though the second is worth a real +0.165 on one arm and we report it as such. What we cannot do without new runs is make the two cells symmetric by construction; a translation-paired corpus, or a Bangla cell with the parenthetical stripped, would settle what we can only bound.

More broadly, this is one corpus, one language, two small open models and one commercial endpoint. The semantics-free control is the only result here that we would expect to transfer as stated; the rest should be read as findings about these systems on this material, and as an instrument other people can point at theirs.

Distractors are drawn from the same corpus, so difficulty is corpus-relative rather than absolute. Jev-1.13 is a commercial endpoint and may change without notice; our results are a snapshot, and we report the exact call count and cost in Appendix R so the snapshot is at least auditable.

Ethics Statement

The corpus is a published, expert-annotated resource of idioms (Sakhawat et al., 2026); it contains no personal data and no content about identifiable individuals. Our interventions rewrite option lists and perturb idiom strings, and produce no new annotation and no generated text.

The work audits two openly released models and one commercial API. We report failure modes in all three, including the vendor’s. We have taken care to separate what is measured from what is inferred. An earlier version of this work reported the commercial arm’s low literal attraction with-

out running it in Bangla; completing the matrix showed that result was a property of the test language, and §5.1 now reports the corrected picture rather than the flattering one. §5.1 also states the pool selection effect against our own cleanest number. Commercial endpoints are mutable; readers should treat the Jev-1.13 results as dated rather than as a standing property of the product.

Bangla is under-served relative to its speaker population, and we think an audit that locates a failure in the language rather than in the task is the more useful kind to publish. The instrument, the per-item outputs and the analysis code are released; Appendix R is the run record they correspond to.

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A The typed question

Every call in this paper sends one typed choice question and receives one probability vector. Nothing else is exchanged: there is no prompt template, no system message, no sampling parameter and no generated text. This is the whole interface, which is why the option set is the only intervention surface available.

Request. For the commercial arm the body posted to `/api/alpha/decisions` is

```
{"model": "typesafe/jev-1.13",
"state": <idiom, or carrier sentence>,
"questions": {"q": {
"type": "choice",
"instructions": <one sentence, Table 5>,
"criteria": {"A": <gloss>, "B": <gloss>,
"C": <gloss>, "D": <gloss>}}}}
```

The two open arms are called locally with the same four fields: the idiom as state, the instruction string, and the four option texts keyed A–D. The option order is shuffled per item by a generator seeded with the item id and the experiment seed, so the same item receives the same permutation in every arm and every family.

Response.

```
{"type": "choice", "choice": "B",
"confidence": 0.75,
"probabilities": {"A": 0.00, "B": 0.84,
"C": 0.16, "D": 0.00}}
```

Arm	Instruction string
Laya-ml	Bangla, see note
Lod-lille	Bangla, identical to above
Jev-1.13	“What is the actual figurative meaning of this idiom?”

Table 5: The instruction is one sentence and is byte-identical across families, experiments and the two Bangla arms. The Bangla string is the direct equivalent of the English one.

We assert on every call that the returned keys are a subset of $\{A, B, C, D\}$ and that the probabilities sum to 1 within 10^{-2} ; no call in any run violated either.

Two confidence signals, and which one we use.

Note that confidence is not the maximum probability: in the example above it is 0.75 while $\max_o p(o) = 0.84$. It is a separate head whose documented job is to estimate whether the correct answer is present in the option list at all. Lod-lille ships the same arrangement. We log it as `conf_head` and report it where it matters (§5.2), but every cross-model number in this paper uses the maximum probability, because it is the only confidence signal all three arms define identically. Where the two disagree the head is not the more diagnostic of the pair: on Lod-lille’s captured items mean head confidence is 0.631 against 0.295 on correct ones, the same inversion the top probability shows.

Answer types we did not use. The interface also accepts `score` (a bounded numeric judgment) and `boolean` questions. We use `choice` exclusively and deliberately. A `score` question would ask the model to rate a gloss in isolation, which destroys the property the instrument depends on: that the gold and the literal reading compete inside a single forward pass, exactly as the psycholinguistic account describes (Cacciari and Tabossi, 1988; Titone and Connine, 1999). A `boolean` question would do the same and additionally reintroduce a yes/no label whose name carries meaning of its own, which is the failure Sun et al. (2026) document. Keeping every experiment in one answer type also means the five distractor families differ in nothing but their option text.

B Positioning against concurrent work

The decision-model literature is a few weeks old at the time of writing, and several papers describe failures adjacent to ours. This appendix states what

each one establishes, what it does not, and why our design is complementary rather than duplicative.

Mechanism versus behavior. Oh et al. (2026) apply causal tracing to autoregressive transformers and recover two pathways carrying an idiom’s figurative and literal readings at once: early layers and specific attention heads promote the figurative interpretation while a direct route keeps the literal one available, and disambiguating context is integrated from the first layer onward. That is mechanistic evidence that the competition our D-LIT condition exploits is real inside a model, and it is what makes our outcome measure interpretable rather than arbitrary: when a decision model places its mass on the literal gloss, there is a documented sense in which both readings were present and one won.

The two designs answer different questions, and the difference is the reason both are needed. Causal tracing requires weight access, token-level activations and a generative forward pass to intervene on. None of the three systems we audit offers the first, two of them emit no token stream to trace, and one is a commercial endpoint behind an HTTP interface. Their method cannot be pointed at the systems that are currently being deployed as routers and guardrails; ours can, at the cost of saying nothing about where inside the model the competition is resolved. They establish *how* the literal route survives; we measure *how often* it takes over, in which language, and with what confidence attached.

Option names and option definitions. Sun et al. (2026) hold a typed model’s option definitions fixed and permute which name is bound to which, and find decision-flip rates up to 70.4 points higher for semantically loaded labels than for neutral ones: the model follows the name rather than the definition it was given. They note that type-safety conceals this, because every output remains structurally valid.

Our o-series (§5.5) manipulates the complementary half. We hold the option names fixed at A–D and destroy the internal structure of the text bound to them, by truncation and by shuffling words within each option, and find accuracy almost unchanged. Read together, the two results bracket the same conclusion from either side: within an option, these models are sensitive to lexical identity and insensitive to composed meaning. Neither result alone would establish that, because theirs is

consistent with a model that reads definitions carefully but weights names more heavily, and ours is consistent with a model that reads definitions as unordered bags.

Context that flips decisions. Xu (2026) append short, fluent, task-preserving context and redirect 61.4% of initially correct decisions, with targeted flip rates of 64.9–73.2% across systems. Our E3 carrier-sentence condition (§5.6) is the benign case of the same mechanism: embedding an idiom in its gold example sentence *raises* accuracy by 0.080 and 0.117 on the two open arms. The two findings are the same sensitivity with opposite signs, and the uncomfortable implication is that a caller cannot tell them apart from the interface. A model whose answer improves when helpful context is appended is, necessarily, a model whose answer moves when context is appended at all.

Calibration and cascades. Ibrahim and Zaki (2026) evaluate decision models against 19 language models on 18 computational social science tasks, and report a median deficit of 11.6 F1 at 44 times lower cost, with confidence calibration *exceeding* most baselines; they recommend decision models as a cheap first stage that routes uncertain items onward. Li et al. (2026) build that system, accepting high-confidence verdicts and escalating the rest, and exceed a frontier model’s accuracy at 41% of its cost.

We reproduce the favourable calibration for the commercial arm on its own language (§5.2), so this is not a contradiction. The boundary we add has two parts. First, in-distribution calibration does not survive a targeted distractor shift, and it fails in the direction that costs a cascade most: on the literal-bearing families the high-confidence tail is where the open arms are *least* accurate, so a confidence gate forwards exactly the items that should have been escalated. Second, and independent of any distribution assumption, the semantics-free control (§5.3) shows what each arm reports when the option list provably carries no information. A gate calibrated on real options has no defense against that case, and a degenerate option list is an ordinary consequence of an upstream bug or an empty retrieval.

What none of this settles. All of the above, ours included, is zero-shot evaluation of base checkpoints. Tang and Zheng (2026) review 28 papers published within nine days of the first widely used

release and conclude that the typed readout has shown no independent accuracy advantage, with the demonstrated gains in latency, cost and cascade placement. Whether any of these failure modes survives task-specific fine-tuning is, as far as we can determine, unmeasured by anyone so far.

C What each experiment changes

Table 7 states, for each experiment, the one thing that varies and the things held fixed. Reading down the “varies” column follows the argument: each row isolates one candidate explanation for the failure measured in E1.

E1, the ladder. D-RAND samples three other idioms’ figurative glosses uniformly. D-SURF samples glosses sharing at least one surface token with the gold. D-SEM samples glosses from idioms sharing at least one semantic tag. D-LIT replaces one sampled distractor with the idiom’s own expert literal gloss. D-HARD applies all three pressures at once. Duplicates are removed and the slate is topped up from the random pool if fewer than k distinct options survive.

E3, perturbing the unit. E3 runs on the literal-bearing option set, so its accuracies are comparable to D-LIT rather than to D-RAND: on the same 1,778 items Laya-ml scores 0.176 under D-LIT and 0.353 under D-RAND, against 0.157 for the intact idiom here. The 0.176 and 0.157 are the same model on the same items under the same family, and they differ because E1 and E3 draw their random distractors from separate seed namespaces (1000 and 3000), so the two option lists share the literal gloss and the gold but not the two sampled distractors. Comparisons are therefore made within an experiment, never across one. v0 is the intact idiom. v1 shuffles its words, v2 reverses them, v3 deletes one word at random, v4 substitutes one word with a word sampled from another idiom, and v5 replaces the idiom with its gold carrier sentence. The option list is held byte-identical across all six, so the only difference reaching the model is the state string.

E6, ablating the options. o0 is the full gloss. o1 truncates every option to its first six words. o2 shuffles the words within each option. o3 replaces each gloss with its counterpart in the other language. o4 replaces every option with a bare label carrying no semantics, which pins accuracy at chance by construction and is the control of §5.3.

D Why interventions on the option set

This appendix explains the design choice the rest of the paper rests on, for readers who want the reasoning rather than the result.

A decision model takes a question and a list of candidate options and returns a probability over exactly those options. It is not autoregressive, it does not sample, and it emits no text. Three of the usual explainability routes are therefore closed. There is no rationale to inspect and no chain of thought whose faithfulness could be measured (Jacovi and Goldberg, 2020; Turpin et al., 2023). Weights are available for two of our arms but not for the commercial one, so no method that requires opening every model, including probing and causal tracing, can be applied uniformly across the three (Oh et al., 2026). And input attribution, the usual fallback (Ribeiro et al., 2016; Lundberg and Lee, 2017), is not only unreliable here (Jain and Wallace, 2019; Adebayo et al., 2018; Hooker et al., 2019) but hard to interpret for this material.

The last point needs stating carefully, because it is what motivates everything else, and the standard shorthand for it is too strong. Idioms are not uniformly non-compositional. Nunberg et al. (1994) argue the opposite of the slogan usually attributed to them: idioms vary in how far their parts can be mapped onto components of the figurative reading, and many are *idiomatically decomposable*, with *spill the beans* distributing its meaning across *spill* and *beans* in a way that *kick the bucket* does not. Psycholinguistic work finds decomposability measurable and consequential for processing (Gibbs et al., 1989; Titone and Connine, 1999).

What follows is weaker than ill-posedness but enough for our purposes. A Shapley value over the tokens of an idiom is always computable, and for a non-decomposable expression there is no reason to expect the resulting per-token shares to correspond to anything linguistically real: the quantity is defined by the attribution game rather than by a decomposition the expression actually supports, and decomposability is a property of the individual idiom that we do not have annotated. The numbers will still arrive, and they will still read as an explanation. We therefore do not build on them, and intervene on the interface instead, which requires no assumption about where an idiom’s meaning sits.

What remains is the interface itself. The caller controls the question and the option list and observes a probability vector. So we intervene there.

option text, as supplied		role	Laya-ml p	Jev-1.13 p
A	একগর্তা; এক সন্তানের জননী; যে নারী একবার মাত্র গর্ভ ধারণ করে	distractor	0.00	0.02
B	ওজন (<i>physical weight</i>)	literal	0.99	0.15
C	ক্ষমতা, গুরুত্ব, মর্যাদা	gold	0.00	0.82
D	হালকাভাবে পড়া	distractor	0.01	0.01

Table 6: **One item exactly as received by the models.** The state is the Bangla idiom ওজন (*ojon*), literally “physical weight,” with the English equivalent *carry weight*. The instruction is identical across items and distractor families. The four option texts are the actual strings reconstructed with the builder and seed used for the run and checked against the recorded gold and literal keys. The probabilities are the returned values. Laya-ml places 0.99 on the literal option, which denotes the physical weight of an object, and 0.00 on the gold figurative gloss; Jev-1.13 places 0.82 on the gold option.

Varies	Fixed	Seed
E1 distractors	idiom, instr., k	1000
E2 idiom surface	meaning, options	2000
E3 idiom form	options, instr.	3000
E4 task language	item, family	4000
E5 nothing (derived)	n/a	n/a
E6 option text	idiom, gold	6000
E7 option order	option set	7000

Table 7: The seven interventions. E5 issues no calls of its own; it is computed from the E1 outputs. Seeds are the Bangla arms’; the commercial arm uses a disjoint set (Appendix R) because its blocks were run in a different order.

The corpus supplies, for every idiom, both a gold figurative gloss and an expert literal gloss, which means a counterfactual distractor is already annotated and we create no new labels: dropping the literal gloss into the option list turns a benchmark item into a targeted probe of whether the expression was read as an idiom or as its words. Holding the item, the instruction and k fixed while varying only the distractor family makes the comparison within-item and within-model, so a difference between two families cannot be explained by item difficulty, by the prompt, or by anything about the model that does not change between the two calls.

The same logic produces the other six interventions. If the failure were about the idiom’s surface form rather than its meaning, E2 would show it (Appendix N). If the model held the expression as a stored unit, breaking the unit in E3 would hurt. If the problem were figurativeness rather than the language, E4 would not move. If the model read options as sentences rather than as bags of words, E6 would collapse under scrambling. If the scorer were order-independent as documented, E7 would return the same gloss under all 24 permutations. Each experiment is an attempt to explain the E1 result away, and the paper reports which attempts succeed.

E The full distractor ladder

Table 8 gives every family for every arm. Two caveats on the commercial column: its numbers are English, and its D-SURF and D-SEM rows run on the larger 8,761-item English pool rather than the 3,265-item literal pool, because those families need no literal gloss.

What the ladder is for. The five families form an ordered difficulty scale in which only the distractor construction changes. D-RAND draws three other idioms’ figurative glosses uniformly, so a wrong answer requires confusing two unrelated meanings. D-SURF draws glosses belonging to idioms that share a word with the target, which makes the distractors lexically adjacent to the state but not to the gold. D-SEM draws from idioms sharing a topical tag, which makes them semantically adjacent to the gold. D-LIT replaces one distractor with the target’s own expert literal gloss. D-HARD combines the literal gloss with one surface and one semantic neighbor. Reading down the first four rows therefore separates three different kinds of hardness, and the point of the ladder is that they are not interchangeable.

The literal family is not merely the hardest. Measured as a drop from D-RAND, the lexical and semantic neighbors cost Laya-ml 0.029 and 0.034, and Lod-lille 0.072 and 0.038. The literal gloss costs them 0.216 and 0.418: roughly six to eleven times as much as either neighbor family, from a single substituted option. A benchmark that stopped at D-SEM, which is where most distractor-hardness work stops, would have recorded these two models as moderately sensitive to distractor difficulty and would have missed the effect entirely. That is the argument for the family being the independent variable rather than a robustness check.

Family	Laya-ml bn	Lod-lille bn	Jev-1.13 en
D-RAND	0.367	0.484	0.968
D-SURF	0.338	0.412	0.915
D-SEM	0.333	0.446	0.829
D-LIT	0.151	0.066	0.931
D-HARD	0.144	0.063	0.854
LAR D-LIT	0.644	0.885	0.047
LAR D-HARD	0.630	0.855	0.043

Table 8: Accuracy by distractor family, and literal attraction on the two families that contain a literal gloss. Chance is 0.250. The commercial column is English and is the reason §5.1 runs the matrix: compare it with the Bangla column of Table 15, where the same model is captured at 0.304.

The commercial column does something different. Jev-1.13’s ordering is not the same shape. Its worst family in English is D-SEM at 0.829, and the literal family at 0.931 is *easier* for it than the semantic one, which inverts the open arms’ ordering. Read alone, that reads as a model which has simply solved the literal distractor. Read against the Bangla column of Table 15, where the same model on the same construction is captured at 0.304, it reads instead as a model for which the literal distractor is easy *in English*. The two readings are empirically distinguishable only because the matrix was run, and we had published the first one before we ran it.

Why D-HARD adds little. D-HARD contains everything D-LIT does plus two harder neighbors, yet costs only a further 0.007 and 0.003 on the open arms. Once the literal gloss is present it absorbs essentially all of the available error: the remaining two slots could be filled with anything without changing the outcome much. This is also why we report LAR on both families and treat D-LIT as the headline, since it is the cleaner single-variable manipulation.

F Selective prediction in detail

Selective prediction asks what happens if the caller answers only the items the model is most confident about. A useful confidence signal makes accuracy rise as coverage falls. Table 9 reports accuracy at low coverage for the two open arms, on the random-distractor condition and on the literal-bearing one.

This is the result with the most direct operational consequence. A cascade that forwards high-confidence answers and escalates the rest (Li et al., 2026; Ibrahim and Zaki, 2026) forwards precisely

	Laya-ml		Lod-lille	
Coverage	D-RAND	D-LIT	D-RAND	D-LIT
0.05	0.669	0.007	0.977	0.000
0.10	0.597	0.007	0.989	0.000
0.20	0.528	0.023	0.899	0.000
0.35	0.470	0.047	0.785	0.000
full	0.367	0.151	0.484	0.066

Table 9: **Accuracy under a confidence gate.** The two model arms are named in the column-group headers and also carry their usual tints, cyan for Laya-ml and magenta for Lod-lille, so the table reads without color. On random distractors, both arms behave as a caller would hope: restricting to the most confident 5% roughly doubles accuracy. On the literal-bearing condition, the curve inverts. Laya-ml falls from 0.151 at full coverage to 0.007 in its most confident 5%, while Lod-lille answers **every one** of its most confident 35% incorrectly. Gating on confidence is therefore worse than not gating, and worse than gating at random.

the items on which these models are wrong, and escalates the ones they would have got right.

How to read the table. Each row fixes a coverage level and reports accuracy on the retained items, so a row is not a subset of the row above it in any model-independent way: the items retained at 0.05 coverage are whichever items *that* model ranked highest, which differ between the two arms and between the two families. The comparison the table supports is therefore vertical within a column, not horizontal across them. The quantity to watch is the sign of the gradient. In the two D-RAND columns it is the sign a caller expects, and the magnitude is substantial: Lod-lille reaches 0.977 in its most confident twentieth against 0.484 at full coverage. In the two D-LIT columns it is reversed, and the reversal is monotone rather than noisy, which is what makes it an inversion of the mechanism rather than a weak signal. The four zeros in the last column are literal: across the 311 items making up Lod-lille’s most confident 35% under D-LIT, the model is correct on none.

Calibration without binning. Table 10 reports the same distributions under three metrics. The point of giving three is that each fails differently. ECE with equal-width bins is the field standard and is also the metric Nixon et al. (2019) show to be sensitive to the binning: with 15 equal-width bins most of the mass can land in one or two bins, and an estimator averaging over near-empty bins can move with the bin count. Replacing them with 15 equal-mass bins, which is their adaptive vari-

Family	acc	ECE	ACE	Brier
<i>Laya-ml</i> ($n = 8,876$)				
D-RAND	0.367	0.071	0.073	0.728
D-LIT	0.151	0.385	0.385	1.030
D-HARD	0.144	0.389	0.389	1.030
<i>Lod-lille</i> ($n = 888$)				
D-RAND	0.484	0.073	0.084	0.608
D-LIT	0.066	0.679	0.681	1.391
D-HARD	0.063	0.673	0.674	1.390
<i>Jev-1.13, English</i> ($n = 3,265$)				
D-RAND	0.968	0.005	0.003	0.048
D-LIT	0.931	0.014	0.013	0.107
D-HARD	0.854	0.047	0.047	0.220

Table 10: **Three calibration metrics on the same recorded distributions.** ■ *Laya-ml*, ■ *Lod-lille*, ■ *Jev-1.13*. ECE uses 15 equal-width bins, ACE the adaptive equal-mass variant of Nixon et al. (2019); the two never differ by more than 0.011, so the ECE figures elsewhere in the paper are not artifacts of the binning. Brier is the multiclass score and needs no bins. Bold marks cells above 0.750, the Brier score of a uniform four-way distribution: in those cells the model’s full reported distribution is less useful to a caller than no preference at all. No new model calls were made; all three metrics are re-readings of the released per-item probabilities.

ant, changes nothing material here: the largest shift across the nine cells is 0.010, and the ordering of the arms is identical. So the calibration failure is a property of the distributions, not of our bin edges.

The Brier score is the stronger statement because it needs no bins at all. It is strictly proper, so no reporting strategy beats reporting one’s true beliefs, and it has a fixed reference point: a model that returns 0.25 on all four options scores exactly 0.750. Under D-RAND both open arms sit below that reference, which is what a model carrying some signal should do. Under D-LIT both rise above it, to 1.030 and 1.391. This is worth stating plainly because it is stronger than miscalibration: on the literal-bearing condition, a caller integrating these models’ full reported distributions is worse off than a caller who replaced them with a flat prior. Jev-1.13 in English stays far below the reference throughout, and the one cell where it approaches a visible calibration cost is D-HARD (0.220, ECE 0.047), which is the condition that also contains the literal gloss.

G Stratified results

Literal capture is not concentrated in rare or obscure idioms, which would be the comfortable explanation. Splitting *Laya-ml*’s D-LIT condition by

the corpus’s own frequency annotation gives LAR 0.671 on *very common* idioms ($n = 1,146$), 0.646 on *common* ($n = 6,200$), 0.615 on *rare* ($n = 1,507$) and 0.565 on *very rare* ($n = 23$). If anything the effect is strongest on the most familiar expressions, which is the opposite of a long-tail story and consistent with these being the idioms whose literal readings are most available.

The gradient is monotone across the three populated strata and spans 0.056, which is small against the effect itself but points the wrong way for a familiarity account. We would not build on the *very rare* cell: at $n = 23$ its interval is wide enough to be consistent with anything, and we report it only so the stratification is complete.

The corpus’s `cultural_significance` field is unusable as a stratifier: 10,337 of 10,359 rows carry the same value. We mention this because it is the field a reader would most want stratified, and its absence is a property of the resource rather than a choice of ours.

Reading Table 11. The table is the one place in the paper where the actual strings reaching the model are shown, and it is worth reading closely because two of its rows look like typesetting errors and are not. The *word shuffle* and *reversed* rows carry the same string: for a three-word expression a particular shuffle and the reversal coincide, and the harness logged both conditions independently and happened to produce the same state. We print what was logged. Equally, the *substitute one word* row replaces the final word with an unrelated token, producing a string that is not Bangla in any idiomatic sense, and the model’s probability on the gold gloss *rises* from 0.21 to 0.24. That single cell is the clearest illustration of §5.6: damaging the expression beyond recognition does not cost the model anything, because the expression was not what it was reading.

The one row that moves is the carrier sentence, and it moves *downward* here ($0.21 \rightarrow 0.07$) even though context raises accuracy on average by 0.080 and 0.117. A single item is not evidence against the aggregate; we include it rather than hunting for an item that agrees with the mean, because the aggregate is the claim and the item is an illustration.

The lower block shows the E6 forms applied to the gold option. The scrambled row is worth comparing word by word with the full gloss: it contains the same lexical inventory in an order no Bangla reader would accept, and costs the models almost

nothing (Table 20). The last row is the semantics-free control, and the table makes concrete what that control removes, which is everything except the option’s label.

H Output distributions

The three arms do not merely differ in accuracy; they place their probability mass very differently, and the shape of that mass is what makes the confidence results of §5.2 possible. Table 12 summarizes the returned vectors.

That last point is worth stating separately, because it is the mechanism behind the inverted reliability curves. A model that found the literal option merely distracting would spread mass and lose accuracy together. Both open arms do the opposite: adding the literal gloss *concentrates* the distribution onto it. On 24 items Laya-ml returns $p = 1.000$ on the literal gloss and $p = 0.000$ on the gold.

How to read the entropy column. Normalized entropy is the entropy of the returned four-way vector divided by $\log 4$, so 1 is uniform and 0 is a point mass. It is reported alongside the maximum because the two can move independently and the pair identifies the shape. A model that is uncertain between two options and has ruled out the other two has a high maximum and a middling entropy; a model hedging across all four has a low maximum and a high entropy. Both open arms under D-RAND sit near 0.89, which is a mild preference over an almost flat vector: at 0.44 mean maximum they are closer to guessing than their accuracy suggests. Under D-LIT Lod-lille’s entropy falls to 0.497 while its accuracy falls to 0.066, so it moves from hedging to committing at precisely the point where committing is wrong.

Jev-1.13 is a different instrument. Its entropy is 0.068 under D-RAND and 85.2% of its answers exceed 0.95, so it is nearly a hard classifier that happens to expose probabilities. That matters for how its calibration figures should be read: a model returning 1.000 almost always has little room to be miscalibrated in the ECE sense while it is accurate, and correspondingly no room to signal doubt when it is not. Its low ECE in Table 3 and its 0.815 confidence on the semantics-free control (§5.3) are the same property seen from two sides.

Worked examples. Table 14 gives five items in the original script, with the annotators’ own

glosses and English equivalents, and Table 13 gives four single-word cases in transliteration for readers who do not read Bengali. Both are drawn from the extreme tail rather than sampled: these are items where the returned mass on the literal gloss is at or near 1.000. The selection is deliberate and we flag it as such, because the tail is what the aggregate LAR of 0.644 does not convey. What the examples add is the *kind* of expression involved. ওজন (*ojon*) is the ordinary word for weight and is used figuratively exactly as English uses *carry weight*; তেল (*tel*) is cooking oil and means flattery or arrogance. These are not literary or archaic constructions, and the literal reading the model selects is in every case the annotators’ own, not one we wrote. The single-word cases are the sharpest, because for them the expression and its literal gloss are the same string, so no amount of attention to the expression’s internal structure could have helped: the only available signal is the task framing, and that is what these models do not use.

I The capture matrix

Table 15 gives the six cells of §5.1 with the accuracy column alongside LAR. All cells are $n = 3,265$ and were built by one function from seed 21000; 95% bootstrap intervals on LAR are [.716, .746], [.363, .397], [.909, .928], [.227, .256], [.288, .319] and [.040, .055] reading down the table. The paired within-model language effect on capture is +0.352 for Laya-ml, +0.677 for Lod-lille and +0.256 for Jev-1.13, all McNemar $p < 10^{-100}$. Jev-1.13’s English cell reproduces LAR from the original run to four decimal places, 0.0472, which is why we treat the two constructions as measuring the same thing. Its accuracy in that cell is 0.926 against 0.931 in Table 8: same model, same items, same family, but the two runs draw their two sampled distractors from different seed namespaces (12000 and 21000), so the option lists share the gold and the literal gloss and differ in the rest. Capture is insensitive to that and accuracy is mildly sensitive, which is what one would expect.

Why this table is the one we defend hardest.

Everywhere else in the paper a comparison moves one thing and holds the rest fixed within one arm. Here the design is a full crossing, so the same 3,265 items, the same distractor draw and the same seed produce all six cells, and both margins are readable. The language margin is large and uniform in sign:

manipulation	string sent to the model	$p(\text{gold})$
E3: the state changes <i>idiom: অগাধ জলের মাছ (Still waters run deep)</i>		
intact	অগাধ জলের মাছ	0.21
word shuffle	মাছ জলের অগাধ	0.19
reversed	মাছ জলের অগাধ	0.19
delete one word	অগাধ জলের	0.24
substitute one word	অগাধ জলের ফ্যা	0.24
in its carrier sentence	সে অগাধ জলের মাছ, কখনো বুঝতে পারি না তার পরিকল্পনা কী।	0.07
E6: the options change <i>gold option rewritten; the other three are rewritten identically</i>		
full gloss	খুব চালাক লোক; সুচতুর ব্যক্তি; যার মনোভাব বোঝা যায় না	
truncated to 6 words	খুব চালাক লোক সুচতুর ব্যক্তি যার	
words scrambled	খুব লোক সুচতুর চালাক যায় ব্যক্তি না যার বোঝা মনোভাব	
English gloss	A very clever person; a shrewd individual; someone whose mindset or intentions are hard to fathom	
semantics removed	A	

Table 11: **What the interventions do to real Bangla text.** The table shows the exact strings used for the idiom অগাধ জলের মাছ (*still waters run deep*, literally “a fish of deep water”). *E3* changes the state supplied to the model; the reported probabilities are those assigned to the gold figurative gloss. Shuffling and reversing the expression leave the probability essentially unchanged, whereas the carrier sentence changes it. *E6* changes the option text while holding its role fixed; the five rows show increasingly reduced or rearranged lexical content, ending with the semantics-free control. The other three options undergo the same transformation as the gold option. The E6 tokenizer splits on punctuation and rejoins on spaces, so the semicolons of the full gloss do not survive into the truncated and scrambled rows; the scrambled row is a strict permutation of the full gloss’s ten tokens and can be checked as one.

		Mean	Med.	>.95	Ent.
Laya	D-RAND	0.438	0.408	0.001	0.892
	D-LIT	0.536	0.479	0.039	0.778
Lod	D-RAND	0.447	0.391	0.002	0.881
	D-LIT	0.745	0.819	0.171	0.497
Jev	D-RAND	0.967	1.000	0.852	0.068
	D-LIT	0.931	0.990	0.708	0.131

Table 12: Distribution of the returned maximum probability, and normalized entropy of the full vector (1 = uniform). The literal-bearing family makes both open arms *sharper*, not flatter: Lod-lille’s mean peak rises from 0.447 to 0.745 and the share of near-deterministic answers from 0.002 to 0.171, while its accuracy falls from 0.484 to 0.066. The model is not confused by the literal gloss; it is convinced by it.

every arm is more captured in Bangla, by +0.352, +0.677 and +0.256. The capability margin is not even monotone. Lod-lille, which beats Laya-ml by +0.133 on the matched random condition (Table 18), is the *most* captured arm in both languages, and the strongest arm is the least captured. A single ordering of these three models therefore cannot predict their capture, while knowing the test language can.

Idiom	Literal	Figurative (gold)	p_{lit}
tel	oil	arrogance	1.000
chamcha	spoon	sycophant	1.000
til	sesame seed	a tiny amount	1.000
charpaya	four-legged	rope cot	1.000

Table 13: Laya-ml under D-LIT. In each case the expert literal gloss and the expert figurative gloss are both in the option list, and the model places all of its mass on the literal reading. These are not hard or obscure idioms; three of the four are single words in everyday use.

Reading accuracy against LAR. The two columns per language are not redundant. Accuracy counts items answered correctly; LAR counts items where the chosen option was specifically the idiom’s own literal gloss, which is one of the three available errors. The gap between them is the residual error on other distractors. For Lod-lille in Bangla, accuracy 0.044 and LAR 0.918 leave 0.038 for everything else, so the literal gloss accounts for almost the whole error budget rather than merely enlarging it. For Jev-1.13 in Bangla, accuracy 0.536 and LAR 0.304 leave 0.160, so even its errors are less literal-dominated. This is the sense in which capture is a specific failure and not a restatement of accuracy.

idiom	expert literal gloss	gold figurative gloss	English equivalent	$p(\text{literal})$
ঈশ্বরপ্রাপ্তি	ঈশ্বরকে অধিগ্রহণ (attainment of God)	মৃত্যু (death)	kick the bucket	0.97
আকাশের চাঁদ	আকাশের চাঁদ (sky's moon)	অজস্র বা দুর্লভ বস্তু; বহু কল্পিত বস্তু (an unattainable or rare object; an imagined object)	pie in the sky	0.99
অল্প জলের মাছ	অল্প জলের মাছ (little water's fish)	সামান্য পুঁজির বা সম্পদের মালিক; অল্প ধন নিযে দরিদ্র ব্যক্তি (a person with limited capital or resources; a poor person with little wealth)	big frog in a small puddle	0.97
ওজন	ওজন (physical weight)	ক্ষমতা, গুরুত্ব, মর্যাদা (power, importance, status)	carry weight	0.99
অগ্নিপরীক্ষা	আগুনের পরীক্ষা (fire test)	কঠিন পরীক্ষা; কঠিন পরিস্থিতির সম্মুখীন (a difficult test; facing a difficult situation)	trial by fire	0.96

Table 14: **Items where Laya-ml places almost all probability on the literal gloss.** Each item contains the Bangla idiom, the expert-provided literal gloss, the gold figurative gloss, and the English equivalent recorded by the annotators. English translations in parentheses are provided for readability; the underlying Bangla strings are the original dataset content. Both literal and figurative glosses were present in the option list. The rightmost column reports the probability assigned to the literal gloss. These are not obscure expressions: *ishwarprapti* corresponds to *kick the bucket* and is read literally as “attainment of God,” while *akashar chand* corresponds to *pie in the sky* and is read literally as “the moon of the sky.”

	Bangla		English	
	acc	LAR	acc	LAR
Laya-ml	0.127	0.731	0.417	0.379
Lod-lille	0.044	0.918	0.684	0.241
Jev-1.13	0.536	0.304	0.926	0.047

Table 15: The literal condition as a model $\times$ language matrix, all six cells measured on the same 3,265 items with one construction and one seed ($n = 3,265$ per cell; 95% CIs in Appendix I). Bold marks the highest LAR per language. Every arm is far more captured in Bangla than in English, and the Bangla ordering does not follow capability: Jev-1.13 is the most accurate and the least captured, while Lod-lille is more accurate than Laya-ml and more captured.

What the table cannot settle. Three things. The pool is the 36.8% of the corpus that carries a usable English gloss, so the levels are pool levels (Appendix J). The two language cells are paired entries rather than translations, and are asymmetric in construction in four ways we enumerate in the same appendix. And the matrix covers the literal condition only; we did not cross the other four families with language, so we cannot say whether the language effect is specific to literal capture or general to distractor difficulty. The first of these affects levels, the second we test directly, and the third is simply unrun.

J How the two language cells were built

The capture matrix is the result we lean on hardest, so this appendix states exactly how its two language cells were constructed, lists every asymmetry between them that we could find, and reports the one test of the largest asymmetry that the existing outputs permit.

Where the glosses come from. Neither literal gloss is a translation produced by us or by a machine. The corpus annotates one field per idiom, `literal_meaning`, written in Bangla with an English parenthetical alongside it, so a single expert-authored field carries both readings. The Bangla cell uses that field as written. The English cell uses the parenthetical alone, recovered by a regular expression that accepts a parenthesized span only if it contains at least three ASCII letters and is at least 60% ASCII alphabetic, which excludes Bangla asides and bare numerals. Of 8,876 items, 3,391 carry such a parenthetical, 3,352 of those also carry an English equivalent idiom, and 3,265 survive a final cut requiring the English literal gloss to be lexically distinct from the English gold gloss (Jaccard < 0.34). That pool of 3,265 is the matrix. The provenance matters because a machine-translated English gloss would make the English cell a measurement of the translator; an annotator-written one does not.

The four asymmetries. We can state these precisely, which is better than asserting the cells are matched.

(i) *The state differs in kind.* The Bangla state is the idiom itself. The English state is the annotator’s nearest English idiom, taken from the `similar_in_english` field, not a translation of the Bangla one. The cells are therefore paired entries, and we use that phrase in the paper rather than “identical items”.

(ii) *The Bangla literal option is bilingual.* Because the Bangla cell uses the annotated field as written, and pool membership requires that field to contain an English parenthetical, every one of the 3,265 Bangla literal options contains its own English gloss inside it, averaging 44 characters against 24 for the English literal option. This is the most serious of the four, because it supplies a cue that has nothing to do with reading the expression: no Bangla figurative gloss in the corpus contains any Latin character, so whenever the literal option carries a parenthetical it is the *only* option in the list written partly in Latin script, and it is the longest. “Select the option containing English” would reproduce the Bangla capture numbers for all three arms and would also predict the English cell showing less capture, because there the cue is absent. §J tests it.

(iii) *The distinctness cut applies to one side only.* The Jaccard < 0.34 filter between gold and literal gloss was applied in English and not in Bangla. Measured after the fact, the filter turns out not to bind: mean gold-versus-literal Jaccard is 0.029 in Bangla against 0.052 in English, so the Bangla side is the more separated of the two and would have passed the cut almost everywhere. This asymmetry is procedural rather than consequential.

(iv) *The literal gloss echoes the state more often in Bangla.* This is the one that could plausibly manufacture the result. The Bangla literal gloss shares at least one content word with the Bangla idiom on 56.2% of pool items; the English literal gloss shares one with the English equivalent idiom on 25.1%. A model doing order-free lexical matching on the option side, which §5.5 shows all three partly are, has a surface hook into the Bangla literal option that it lacks in English.

Testing asymmetry (iv). The existing per-item outputs are enough to test it without running anything new. Split the pool on whether the Bangla literal gloss shares a content word with the idiom and

	Laya	Lod	Jev
<i>Bangla LAR by surface hook</i>			
hook absent ($n = 1429$)	0.716	0.845	0.376
hook present ($n = 1836$)	0.742	0.975	0.247
difference	+0.026	+0.130	−0.129
<i>Capture gap, hook-absent items only</i>			
Bangla LAR	0.716	0.845	0.376
English LAR	0.358	0.236	0.066
gap	+0.358	+0.610	+0.310
headline gap (all pool)	+0.352	+0.677	+0.256

Table 16: **The construction asymmetry does not carry the result.** Laya-ml, Lod-lille, Jev-1.13. “Hook” means the Bangla literal gloss shares at least one content word with the idiom, which is the asymmetry most likely to inflate Bangla capture. Upper block: capture is not reliably higher when the hook is present, and on the commercial arm it is lower. Lower block: restricting both language cells to hook-absent items leaves the capture gap essentially unchanged, and larger for two of three arms. The script is released with the per-item outputs.

read LAR in each stratum (Table 16, upper block). If the hook were driving capture, the overlap stratum would be far more captured in every arm. It is not: the effect is +0.026 for Laya-ml, +0.130 for Lod-lille and −0.129 for Jev-1.13. The sign is not even consistent across arms, and on the commercial arm the hook is associated with *less* capture, which is the opposite of the predicted direction.

The sharper test restricts both language cells to the 1,429 items where the hook is absent and re-reads the gap there (lower block). It barely moves: +0.358, +0.610 and +0.310, against headline gaps of +0.352, +0.677 and +0.256. For two of the three arms the gap is slightly *larger* once the hook is removed. We read this as showing that asymmetry (iv) does not account for the Bangla–English difference, not as showing the cells are equivalent; a corpus paired by translation rather than by annotator judgment would settle what we can only bound here.

Testing asymmetry (ii). The capture matrix cannot test this, because every item in its pool carries a parenthetical by construction. The Bangla arm of E1 can. It runs the D-LIT family over all 8,876 items, and on 5,485 of them the literal gloss carries no parenthetical at all, so the Latin-script cue is simply absent. If the cue were what produces capture, those items should show little of it.

They show a great deal of it (Table 17). Two things follow, and they point in opposite directions.

	Laya	Lod
<i>Bangla LAR under D-LIT</i>		
cue absent	0.581	0.871
<i>n</i>	5,485	560
cue present	0.746	0.909
<i>n</i>	3,391	328
difference	+0.165	+0.037
95% CI	[.145, .184]	[−.005, .077]
<i>same split, families with no literal option</i>		
D-RAND	+0.014	−0.028
D-SURF	+0.025	−0.020
D-SEM	+0.030	+0.042

Table 17: **The Latin-script cue is real but does not carry the result.** Laya-ml, Lod-lille. Items are split by whether the idiom’s literal gloss carries an English parenthetical, which is the only thing in a Bangla option list written in Latin script. Upper block: the cue is worth +0.165 of capture on the marker scorer and little on the independent-option scorer, yet capture without it is still 0.581 and 0.871 against a chance level of 0.250. Lower block: the same split barely moves accuracy on families containing no literal option, so it is not a proxy for item difficulty. The commercial arm cannot be split this way, since its only Bangla literal run is the pool, where every item carries a parenthetical.

First, the mechanism is real, at least on the marker-based scorer. Laya-ml’s LAR is 0.746 where the cue is present against 0.581 where it is absent, a difference of +0.165 with a bootstrap interval of [+0.145, +0.184]; on Lod-lille it is +0.037 with an interval straddling zero. Part of the Bangla capture we report is therefore attributable to a cue rather than to a reading of the expression, and we would not have found that without looking.

Second, it is not what produces the result. With the cue entirely absent, Laya-ml still selects the literal gloss on 58.1% of items and Lod-lille on 87.1%, against a chance level of 0.250 and against English-cell capture of 0.379 and 0.241. The effect is also specific: on the three families that contain no literal option, the same split moves accuracy by at most 0.042 with inconsistent sign, so the parenthetical stratum is not simply the easier or harder half of the corpus.

This bounds the cue rather than removing it. It covers the two open arms and not the commercial one, and the no-cue stratum is a different item set from the pool, so these numbers are not a clean subtraction from the matrix. The experiment that would settle it is to strip the parenthetical from the Bangla literal option and re-run the Bangla cell, which we have not done.

What this appendix does not establish. Asymmetry (i) remains untested: we cannot ask what the arms would do on an English state that is a literal translation rather than an idiomatic equivalent without running that condition. We flag it in Limitations rather than dismissing it.

K Cross-lingual arms

Figure 7 plots the E4 conditions discussed in §5.7. E4 differs from the capture matrix in two ways that are easy to conflate, so we state them. E4 uses the D-RAND family, with no literal gloss anywhere in the option list, and it asks what the language costs on an ordinary benchmark item. The matrix uses D-LIT and asks what the language does to literal capture specifically. The two therefore measure different things and the figure should not be read as a version of Table 15.

What the four Laya-ml bars are. Laya-ml has two checkpoints in play. The first two bars are the multilingual checkpoint on Bangla and on English, which is the within-model language contrast. The third and fourth use the vendor’s English-only checkpoint, and exist as a control rather than as a result: the third runs it on the English condition, where it should and does perform, and the fourth runs it on Bengali script, where it sits at 0.283, barely above the 0.250 chance line. That fourth bar is doing one specific job. It rules out the possibility that our Bangla numbers are an artifact of the option-marker scoring harness rather than of the language, because a checkpoint that is documented as unable to read the script fails here in the documented way. If the harness were manufacturing the Bangla deficit, this bar would not be distinguishable from the multilingual checkpoint’s Bangla bar, and it is.

The gains are not uniform, and the ordering is informative. Lod-lille gains the most (+0.380), Laya-ml and Jev-1.13 roughly similar amounts (+0.246 and +0.230). Lod-lille is the arm whose documentation states English-only training evaluation, and it is also the arm whose Bangla accuracy (0.484) is closest to chance relative to its English accuracy (0.864). The natural reading is that its figurative knowledge is largely intact and the Bangla interface is where it is lost. We cannot confirm that without probing the model, which we do not do here, and we note the alternative: a model may also be worse at Bangla figurative semantics specif-

Model	n	Accuracy	95% CI
Laya-ml	8,876	0.367	[0.357, 0.377]
Lod-lille	888	0.484	[0.453, 0.518]
Jev-1.13	8,761	0.736	[0.727, 0.745]

Table 18: **The common baseline across all three arms.** The condition is identical across models: a Bangla idiom followed by its Bangla figurative gloss with random distractors. All pairwise McNemar tests on the overlapping items are significant: Jev–Laya +0.369, Jev–Lod +0.259, and Lod–Laya +0.133 (all $p < 10^{-9}$).

ically, which would produce the same external behavior.

Why the error bars matter here. The three arms run at very different n , so the intervals differ by a factor of three in width. Lod-lille’s English condition in particular runs on its 10% subset. All three gaps clear McNemar at $p < 0.001$ on paired items, which is the test to read rather than the overlap of the bars, since the bars are unpaired marginals and a paired effect can be significant while marginal intervals overlap.

L The matched three-way condition

Figure 8 plots Table 18, the one condition that is identical across all three arms: Bangla idiom, Bangla glosses, random distractors, four options, gold position randomized.

Why only one condition qualifies. Every other comparison in the paper is scoped by something the three arms do not share. The literal condition needs an annotated literal gloss, which exists in Bangla for the whole corpus but in English only for the 3,265-item pool. The confidence results need a comparable confidence signal, and the three arms expose three different ones: Laya-ml’s fitted per-bucket temperatures, Lod-lille’s separate confidence head, and Jev-1.13’s raw maximum. The order experiment needs the full permutation set, which we ran at different n per arm. This condition needs none of that, so it is the one place where a bare capability ordering can be read without a caveat attached.

What the ordering is for. It exists to be contradicted later. The ordering is Jev-1.13 0.736, Lod-lille 0.484, Laya-ml 0.367, with every pairwise McNemar significant at $p < 10^{-9}$, so it is not a close call. Three later results do not follow it. Capture is highest on the middle arm (Table 15). Calibration is best on the strongest arm by ECE and

worst on the strongest arm on the semantics-free control, where it reports 0.815 confidence at 0.254 accuracy. Order stability is highest on the two arms with the independent-option and commercial designs, which are the strongest and the middle arm, and lowest on the weakest. Without a baseline ordering none of those non-correspondences would be storable, which is why a table this simple earns its own appendix.

The one caveat. Jev-1.13’s cell runs on 8,761 items against Lod-lille’s 888, and the three arms are not the same size, architecture or license. The condition is matched on the *task*, not on the models, and a capability ordering across three systems chosen for architectural contrast is not a benchmark ranking. We use it as a reference point internal to this paper and would not quote it as one.

M Unit integrity

Figure 9 plots the E3 perturbations described in §5.6. The options are byte-identical across all six variants, so the only difference reaching the model is the state string. Table 11 shows the six states for one idiom exactly as they were logged.

What each perturbation is supposed to break.

The six states form a graded attack on the expression’s integrity. *Intact* is the baseline. *Word shuffle* destroys word order while preserving the lexical inventory, so a model storing the expression as an ordered unit should fail while a model matching its words should not. *Reversed* is the specific shuffle that is maximally far from the original in order while being deterministic rather than sampled. *Delete one word* removes material, which should hurt any model relying on the full form. *Substitute one word* replaces material with an unrelated token, producing a string that is not the idiom at all and is not idiomatic Bangla. *Carrier sentence* adds the corpus’s own example sentence around the intact idiom, which adds disambiguating context without touching the expression.

Reading the two panels together. Accuracy on the left and intact-agreement on the right answer different questions, and either alone is misleading. Accuracy can stay flat because the perturbation does nothing, or because it changes many answers in compensating directions. Agreement distinguishes those: it counts the share of items whose answer is unchanged from the intact state, so a model that is unaffected scores high on both

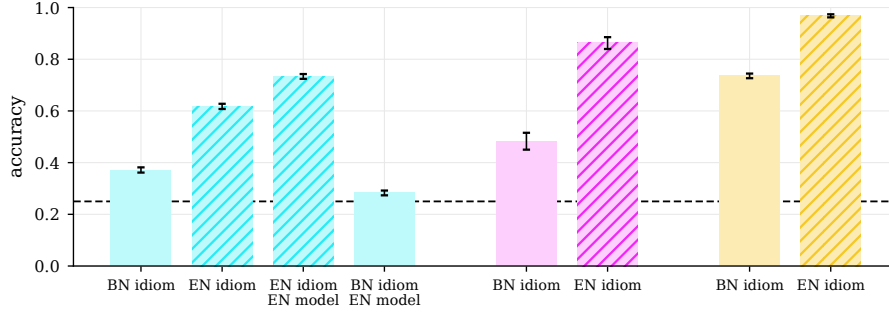


Figure 7: Accuracy by task language, grouped by model: Laya-ml (four bars), Lod-lille, Jev-1.13. Solid bars are Bangla input, hatched bars English; Laya-ml’s third and fourth bars use the English-only checkpoint, the fourth being the control. Error bars 95% bootstrap CIs, dashed line chance. Every model gains between 23 and 38 points when the paired English entry is presented instead of the Bangla idiom.

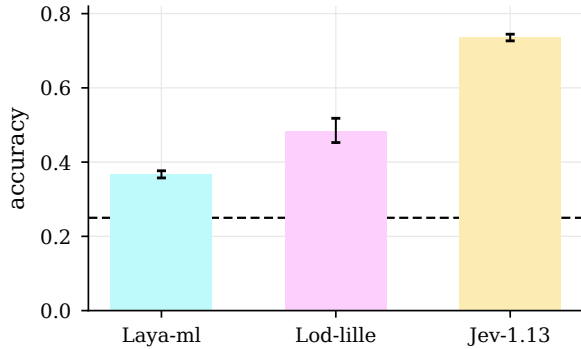


Figure 8: The matched three-way comparison. Bars left to right: Laya-ml, Lod-lille, Jev-1.13. Black dashed line is chance, error bars 95% bootstrap CIs. This capability ordering is the baseline against which the calibration failures in §5.2–§5.3 should be read.

while a model that is churning scores flat accuracy and low agreement. Laya-ml agrees with itself on 0.742 of items under shuffling and Lod-lille on 0.917, so neither is churning. The conclusion is the stronger one: the perturbation is close to invisible to both, not merely harmless on average.

What this rules out, and what it does not. It rules out the hypothesis that these models hold the idiom as a stored lexical unit whose retrieval is keyed on its exact form, because breaking the form does not cost them the answer. It does not establish what they do instead. In particular, flat accuracy under shuffling is also consistent with a model that was never using the state string for much at all, which the carrier-sentence row supports: that is the only manipulation that moves either arm, by +0.080 and +0.117, and it is the one that adds context rather than altering the expression. Our reading is that the usable signal is in the surrounding material rather than in the expression, and we state it as a reading.

Power. Lod-lille’s E3 runs on 180 items, so several of its individual perturbation contrasts are not

separately significant and we do not claim them. The claim is the pattern across six conditions and two arms, and the one contrast that is individually large in both.

N Surface invariance

The corpus ships alternative surface forms certified by annotators as meaning-preserving: a variant spelling, a clipped form, or a near-synonymous rewording recorded in the `alternative_idioms` field by the same experts who wrote the glosses. These are not perturbations of ours. A model reading meaning rather than form should return the same answer on both members of such a pair. E2 presents each pair under the `D-HARD` family, so the literal gloss is in the option list for both members and the comparison is made under the condition where capture is strongest.

Laya-ml agrees with itself on 0.657 of 3,143 pairs (2,423 items) and Lod-lille on 0.677 of 310 pairs (234 items): about a third of the time, a rewrite the annotators certified as meaning-preserving changes the answer. Mean Jensen–Shannon divergence between the two returned probability vectors is 0.039 and 0.085, on a scale where 0 is identical and 1 is disjoint support.

Reading agreement against divergence. The two numbers look inconsistent and are not. Small mean divergence with frequent answer changes is the signature of decisions taken on thin margins: the rewrite moves the distribution only slightly, but enough to cross the argmax boundary. Laya-ml’s mean peak probability on these pairs is 0.535, so its top two options are often close and a small shift reorders them (Appendix H gives the full distributions). Lod-lille is the harder case at 0.737: its distributions are sharp, yet it still changes answer on a third of pairs, so for that arm the thin-margin read-

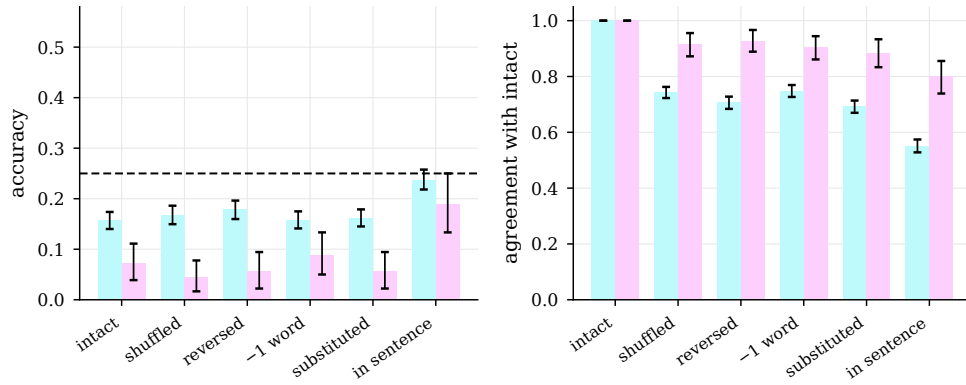


Figure 9: Unit integrity: the idiom is perturbed, the options held fixed. ■ Laya-ml, ■ Lod-lille; error bars 95% bootstrap CIs, dashed line chance (left panel). *Left*: accuracy. *Right*: agreement with the answer on the intact idiom. Scrambling and reversing the expression does not meaningfully hurt either model, which is the bag-of-words signature.

ing covers part of the instability rather than all of it.

Why this is reported as a control rather than a headline. E2 is the experiment designed to explain the E1 result away, and it does not. If capture were an artifact of the particular surface form the corpus happens to record, presenting the annotators’ alternative form should recover the figurative reading. It does not. Accuracy does move, and significantly: $0.138 \rightarrow 0.193$ for Laya-ml and $0.035 \rightarrow 0.158$ for Lod-lille (McNemar exact $p < 0.001$ on both), so the surface form is not irrelevant. These canonical-form accuracies are lower than the D-HARD rows of Table 8 (0.144 and 0.063) because E2 runs only on the items that carry an annotated alternative form, 2,423 of 8,876 and 234 of 888, and that subset is harder for both arms; the comparison here is within the pair, so the level does not affect it. But both forms sit far below the 0.250 chance level on both arms, so the alternative form rescues neither model. We read the two results together: these scorers are genuinely unstable under meaning-preserving rewriting, which is worth reporting on its own, and that instability is neither large enough nor in the right direction to be the mechanism behind a LAR of 0.644 to 0.885.

Caveat on the pairs. Items vary in how many alternative forms they carry, so the pair counts exceed the item counts and items with more variants are weighted more heavily. We report the pair-level figure because it is the quantity a caller experiences; averaging within item first gives 0.667 and 0.697, so the weighting raises agreement by 1.0 and 1.9 points and the pair-level figure is the more conservative of the two.

O Subset representativeness

Lod-lille ran a fixed 10% random subset, drawn once with seed 909 and reused for every experiment on that arm, so its cells are comparable to each other. The question this appendix answers is whether they are comparable to the other arm’s full-set cells. Because the subset is a subset of the same corpus, we can measure the answer directly rather than assume it: Table 19 runs Laya-ml, which has the full set, on those same 888 items and compares it with itself.

Why this is the right test. An alternative would be to compare Lod-lille on the subset against Laya-ml on the full set and attribute any difference to sampling, which conflates the sample with the model. Holding the model fixed and varying only the sample isolates the sampling term. Whatever drift appears in Table 19 is an upper estimate of what the subset contributes to a cross-arm comparison, since the two arms see the same items.

What the drift column shows. The largest absolute drift across five families is 0.016, on D-RAND, and the signs are mixed rather than systematic: three families drift down, two up. A subset that over-represented easy or hard items would drift consistently in one direction. Two of the five subset estimates fall marginally outside the full-set bootstrap interval, which at $n = 888$ against $n = 8,876$ is the expected rate rather than a signal. Critically, no family’s drift is large enough to reorder the ladder: D-LIT and D-HARD remain far below the other three on both the full set and the subset, and the D-RAND to D-LIT gap, which is the quantity the paper reports, changes by 0.001.

Family	Full	Subset	Drift
D-RAND	0.367	0.351	−0.016
D-SURF	0.338	0.339	+0.001
D-SEM	0.333	0.325	−0.007
D-LIT	0.151	0.136	−0.015
D-HARD	0.144	0.149	+0.004

Table 19: Laya-ml on the 10% subset against its own full set. Maximum drift is 0.016 and the signs are mixed, so there is no systematic easy-or-hard skew. Two of five subset estimates fall marginally outside the full-set bootstrap CI, which is expected at $n = 888$ and changes no sign or ordering. The full-set figures are therefore reported as the precise estimates, and Lod-lille’s intervals are reported wider throughout.

P Full option ablation

Table 20 gives the numbers behind Figure 10. The five rows hold the state, the instruction and the gold position fixed and rewrite every option in the list the same way, so a row is a statement about what the option text needs to contain for the decision to survive.

The first three rows are the finding. Truncating each option to its first six words costs 0.004, 0.024 and 0.011 across the three arms. Scrambling the words *inside* each option, which leaves a string no reader of either language would accept as a sentence, costs 0.008, 0.013 and nothing at all; on Jev-1.13 the scrambled row is marginally *above* the full gloss, which is within noise but makes the point that the manipulation is invisible to it. Taken together: what these models need from an option is its lexical inventory, and the order and syntax of that inventory are close to free. This is the behavioral half of the result Sun et al. (2026) reach from the other direction by permuting which option name is bound to which definition.

The fourth row is a separate effect. Presenting the options in the other language, with the state unchanged, costs Laya-ml 0.018 and Jev-1.13 0.007 but Lod-lille 0.095, which is five to thirteen times as much. The arm that loses most from a language mismatch between state and options is the arm whose documentation reports English-only training evaluation, and the one with the largest cross-lingual gap in E4. We note the correspondence; with three models we cannot test it.

The fifth row is the control, and it is the row that validates the instrument. Replacing every option with a bare label leaves 0.256, 0.233 and 0.254, against a chance level of 0.250. This mat-

Option text	Laya	Lod	Jev
full gloss	0.365	0.492	0.970
truncated	0.361	0.468	0.958
scrambled	0.357	0.479	0.974
other lang.	0.347	0.398	0.963
no semantics	0.256	0.233	0.254

Table 20: **Option-side ablation.** The semantics-free row removes meaningful option content and yields near-chance accuracy in all three arms. Jev-1.13’s column is English.

ters more than it looks. It shows that the four preceding rows are measuring option semantics rather than some artifact of the harness, because when the semantics are removed the measurement collapses to exactly where it should. It also shows the gold-position randomization is working: a harness with a position leak would score above chance here. And it is the condition §5.3 builds on, because it is the one place where accuracy is known a priori and any confidence above zero is therefore unambiguously excess.

Caveats. Jev-1.13’s column is English throughout, so its rows are not comparable in level to the other two, only in shape. The three arms’ *full gloss* rows differ slightly from the corresponding D-RAND rows of Table 8 because E6 draws its own distractor sample under seed 6000; the differences are 0.002 to 0.008 and we report both rather than reconciling them silently.

Q Order sensitivity and position bias

This appendix reports the per-slot selection shares behind §5.4, then sets out the permutation audit that establishes the residual instability of the independent-option scorer is real rather than an artifact of our harness.

What is held fixed. Each item here is one idiom with one fixed set of four option texts. We present that same set 24 times, once per permutation, and change nothing else: the instruction string, the state, k , the decoding configuration and the option texts themselves are byte-identical across the 24 calls, and no call is retried. The only quantity that varies is which slot each option occupies. Anything that moves across the 24 calls is therefore attributable to order, which is what makes this the cleanest of our seven interventions.

Recovering option identity. The harness records, per call, the permutation as a four-digit string in which position s holds the index, in the

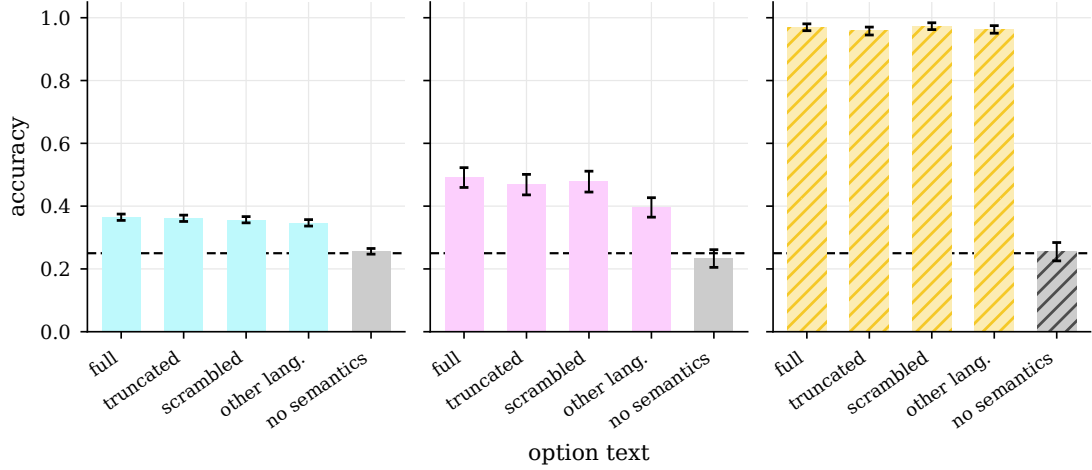


Figure 10: Option-side ablation. Panels left to right: Laya-ml, Lod-lille, Jev-1.13. In each panel the fifth, hatched bar is the semantics-free control of §5.3; the whole Jev-1.13 panel is hatched because its task is in **English**. Error bars are 95% bootstrap CIs and the black dashed line is chance. Truncating and word-scrambling the option text cost almost nothing in any of the three models.

canonical option list, of the option displayed in slot s . A chosen slot must be mapped back through that permutation before it can be compared across calls. Counting distinct slot *letters* instead inflates instability to near unity for any scorer, because a letter necessarily changes whenever the same gloss is moved, and we report this because an earlier draft of our own analysis made that mistake. All numbers in the paper use the mapped gloss identity.

The permutation audit. Table 21 reports a quantity that the summary measures in Table 4 cannot: not whether the *answer* changes, but how far a *single option's own score* travels as that option is moved around the list. This is the quantity the independent-option design makes a claim about. Scoring each option in its own forward pass should make an option's score a function of the option and the state alone, so the spread should sit at the level of floating-point accumulation noise, which for fp32 summation over sequences of this length is on the order of 10^{-7} . It does not. Lod-lille's median item has an option whose score moves 4.0×10^{-2} , five orders of magnitude higher, and not one of its 299 items falls below 10^{-4} . Two consequences follow. First, the 10.7% answer-change rate in Table 4 is not a tie-breaking or slot-indexing artifact of our harness: ties would require near-identical scores, and the scores are not near-identical, so what moves is the scoring itself. Second, the documented guarantee is not one an evaluation can rely on, at least in the configuration we were able to run. We take the narrow reading throughout: the independent-option design is

	Laya	Lod	Jev
items	992	299	300
median spread	0.495	0.040	0.070
mean spread	0.507	0.046	0.139
95th pct. spread	0.772	0.105	0.480
max spread	0.928	0.245	0.820
share $< 10^{-4}$	0.000	0.000	0.230
answer changes	0.837	0.107	0.133

Table 21: **Permutation audit.** Per item, the largest distance a single option's own score travels across the 24 orderings of its list, after mapping each slot back to the option it displayed. Laya-ml, Lod-lille, Jev-1.13. Bold marks the largest value in each row. An order-independent scorer should sit at fp32 noise, around 10^{-7} ; the *share* row gives the fraction of items at or below 10^{-4} , a threshold three orders of magnitude more permissive than that. Only Jev-1.13 has any items at that threshold, and then fewer than a quarter; its probabilities are reported to two decimal places, so for that arm the row means identical *as reported* rather than identical in the underlying computation. The script that produces this table is released with the per-item outputs.

a large improvement over the marker-based one on this axis, and it is not the categorical property its documentation describes.

The commercial arm. Jev-1.13 makes no public invariance claim, and its behavior is the least like either open arm. It returns identical distributions on 23.0% of items, with the caveat above that its probabilities reach us rounded to two decimal places, so this describes the reported values rather than the computation behind them. Against that, its mean spread (0.139) is three times Lod-lille's and its maximum (0.820) approaches Laya-ml's. The distribution is bimodal rather than merely noisier,

which is what one would expect from a scorer that is close to deterministic on easy items and switches regime on hard ones. We cannot test that reading without weights, and we do not claim it.

Position bias. Figure 12 gives the distributional summaries behind Table 4, both from the permutation sweep. Figure 11 is the one panel in this appendix drawn from a different run: it reports per-slot selection shares over the full ladder condition, all 8,876 items for Laya-ml and 8,761 for Jev-1.13, because position is randomized per item there and the estimate is far better powered than the sweep’s 992 items would allow. The sweep agrees in direction and is milder, 32.7% against 18.8% on slot B and slot A for Laya-ml. Because the gold option is placed uniformly at random in both, the expectation under no positional bias is exactly 0.25 in every slot for every arm, so neither needs a baseline model: the distance from the dashed line *is* the bias. Laya-ml’s profile is not the primacy gradient usually reported for prompted generative scorers (Zheng et al., 2024); it peaks on slot B at 37.2% against 14.6% for slot A, a $2.5\times$ swing in selection share from position alone, and is not a monotone function of position at all. We note the shape and decline to explain it; with one model per scoring design, we have no way to separate a property of the marker scheme from a property of this particular backbone.

R Run record

Environment. Python 3.13.15 throughout. The two open arms ran on one Tesla T4 (15,360 MiB): Laya-ml in bf16 with `max_len` 1024 at 27.2 ms per call, Lod-lille in fp32 at 86.1 ms per call. The commercial arm needs no accelerator and ran on CPU against `typesafe/jev-1.13`.

Temperatures. Laya-ml’s run recorded `temperature = [1.0, 1.0, 1.0]` with `temperature_by_options` empty, so the per-bucket temperatures it ships were not applied and its ECE figures are raw. Lod-lille ships `temperature: 1.0` by design, its authors having reported that fitting hurt held-out calibration. Both open arms are therefore untempered and comparable on that axis. The commercial arm’s internal calibration is not observable.

Seeds. The Bangla arms use a per-experiment seed (E1 1000, E2 2000, E3 3000, E4 4000, E6 6000, E7 7000) added to the item id, so each item’s

option permutation is reproducible and identical across arms. The split seed is 20261001. The commercial arm uses a disjoint block seed set (order audit 17000, literal 12000, ceiling 11000, Bengali control 14000, options 16000) because its blocks ran in a different order; the permutation for a given item is still a pure function of seed and id.

Call volume and cost. The commercial arm issued 56,533 calls for \$0.9348 at \$0.042 per million input tokens with output free. The blocks are the order audit (300 items $\times$ 24 permutations = 7,200), literal capture on the shared pool ($3,265 \times 3$ families, D-RAND, D-LIT and D-HARD, = 9,795), the English ceiling ($8,761 \times 3$ families, D-RAND, D-SURF and D-SEM, = 26,283), the Bengali control (8,761, one family) and the option ablation (873×5 option forms = 4,365), which is 56,404 retained rows. The option ablation runs on 873 items rather than the 888 of the shared subset because this block is English and 15 of the 888 carry no English equivalent idiom to use as the state; the 873 are exactly the intersection of the subset with the 8,761 items that do. The remaining 129 calls are the endpoint probe and the response-contract checks made before each block. Transient 429 and 5xx responses were retried with exponential backoff to a 30 s ceiling; no call was dropped.

Statistics. All intervals are 95% percentile bootstrap over items, 2,000 resamples, fixed seed 0. ECE uses 15 equal-width bins. Paired comparisons use McNemar’s exact test via the binomial distribution. AURC is reported against a random-gate baseline computed on the same items.

What is released. The per-item outputs for every cell of every experiment (one row per item per condition, carrying the full probability vector, the predicted key, the gold key, the literal key and the gold position), the five split manifests, the three run environment records quoted above, the task configurations containing the verbatim instruction strings, the analysis scripts that turn those outputs into every table in this paper, the figure sources, and, beside each figure, a CSV of the exact plotted values.

Pairing. Because the two open arms were reconstructed independently from a written contract rather than from shared code, the released outputs admit a direct check: on the 4,440 overlapping rows the gold key, gold position, idiom string and literal-distractor key agree at 1.000. That check is

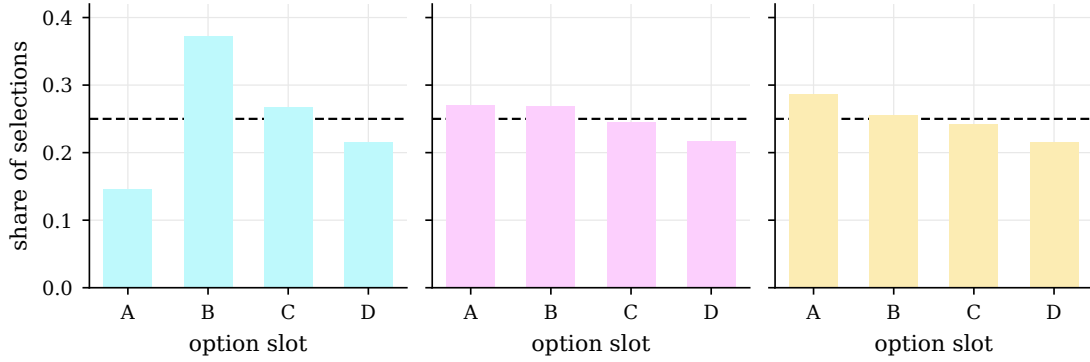


Figure 11: Share of selections falling on each option slot, over the full ladder condition rather than the permutation sweep. Panels left to right: Laya-ml, Lod-lille, Jev-1.13. The gold option is placed uniformly at random, so the black dashed line at 0.25 is the expectation under no positional bias and any departure from it is bias. Laya-ml peaks on slot B, taking it 37.2% of the time against 14.6% for slot A; the other two arms sit close to the line throughout, with a mild and monotone preference for earlier slots.

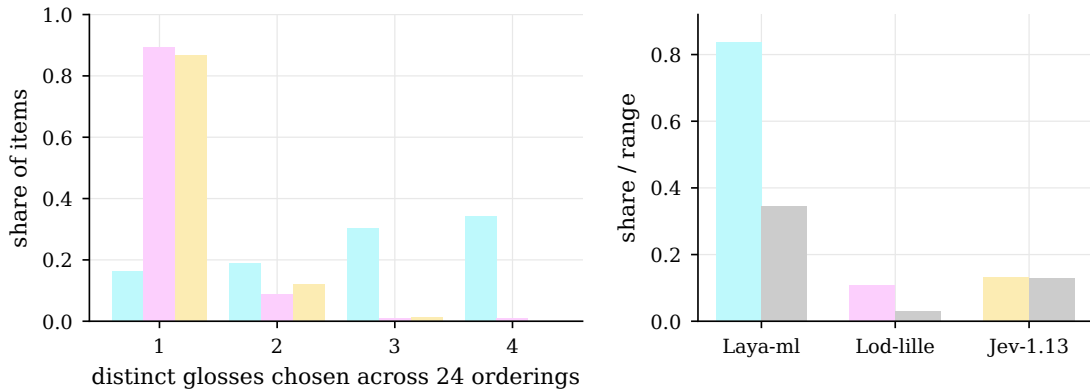


Figure 12: Order sensitivity. *Left*: how many distinct glosses each model selects across the 24 orderings of one option set; within each group bars run Laya-ml, Lod-lille, Jev-1.13, and a value of 1 means perfect stability. *Right*: per model, the colored bar is the share of items whose chosen gloss is unstable and the grey bar is the mean within-item range of $p(\text{gold})$. Laya-ml is unstable on 83.7% of items with a mean $p(\text{gold})$ swing of 0.346; the other two are near-stable.

itself a released artifact, and it is what licenses every paired test in the paper.

The matrix run. The six cells of §5.1 were run afterwards as one job, with its own seed namespace (21000) so no option permutation collides with the runs above. Every cell covers the same 3,265 items and is built by a single function; the two languages share one distractor draw per item and are shuffled independently from the same generator. Laya-ml ran both languages on the multilingual checkpoint, so only the language changes and the checkpoint does not. Lod-lille ran the full pool rather than its earlier 10% subset, which supersedes the thin 316-item in-pool estimate. The commercial arm issued 6,531 calls for \$0.1152. Its English cell reproduces the value from the original run to four decimal places (0.0472), which is the check that the two constructions measure the same thing.

Known gaps. E2 was not run on the commercial arm, because the alternative surface forms exist only in Bangla and the arm’s task is English. The matrix covers the literal condition only; the other distractor families were not run in both languages. The confidence and calibration results remain two-arm, because the commercial arm’s reliability was measured on its English run.